INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Machine Learning Techniques (MLT)

Past Year Question Bank (44 Exam Sessions) — Official Categorized Syllabus Edition

Total Unique Questions: 462	**Exam Sessions Covered:** 44 Papers	**Format:** MCQ, MSQ, NAT / SA
Week 1: PCA & Intro (343 Qs)	**Week 2:** Kernel PCA (8 Qs)	**Week 3:** Clustering - K-means (72 Qs)
Week 4: Estimation & GMM (39 Qs)	**Fidelity:** Official TCS iON Cards	**Interactive Mode:** NTA JEE CBT Online

Week 1: Introduction; Unsupervised Learning - Representation learning - PCA (343 Questions)

• PCA Fundamentals & Projection (302 Questions)

MLT_001 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653345305 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT " MACHINE LEARNING TECHNIQUES "

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531148552. ✔ YES

6406531148553. ✖ NO

✔ Official Answer Key: Option (A)

MLT_002 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 640653345306 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

Note:

Do not write your answer as percentage. Always enter fractions as they are. e.g. if your answer is 0.245, enter the same, do not enter 24.5 %.

Options :

6406531148554. ✔ Useful Data has been mentioned above.

6406531148555. ✖ This data attachment is just for a reference & not for an evaluation.

Sub-Section Number :	2
Sub-Section Id :	64065348795
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_003 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653345307 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

For the given data set

x	y
1	3
2	4
3	8
4	11
5	12

What is the mean squared error if the predicted model is given by $\hat{y} = 2x + 1$? Write your answer correct upto 2 decimal places, do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.38 to 1.42

Sub-Section Number :	3
Sub-Section Id :	64065348796
Question Shuffling Allowed :	No

Question Id : 640653345308 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (173 to 176)

✔ Official Answer Key (Numerical Value): 1.38 to 1.42

MLT_012 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 181 Question Id : 640653345314 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the output of following code snippet?

```
print(C[1][0]+D[1][2])
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ **Official Answer Key (Numerical Value): 5**

MLT_015 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 184 Question Id : 640653345328 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What do you think will happen if Abhishek tries to fit a linear regression model with polynomial transformation of degree 20?

Options :

6406531148590. ✖ High validation loss and high training loss.

6406531148591. ✔ High validation loss and low training loss.

6406531148592. ✖ Low validation loss and high training loss.

6406531148593. ✖ Low validation loss and low training loss.

Sub-Section Number :	5
Sub-Section Id :	64065348798
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_017 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 186 Question Id : 640653345317 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

The gradient of function $f(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2$ at point $(x_1, x_2) = (1, 2)$ is:

Options :

6406531148567. ✔ $[4, 5]^T$

6406531148568. ✖ $[5, 4]^T$

6406531148569. ✖ $[2, 3]^T$

6406531148570. ✖ $[4, 6]^T$

✔ Official Answer Key: Option (A)

MLT_018 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

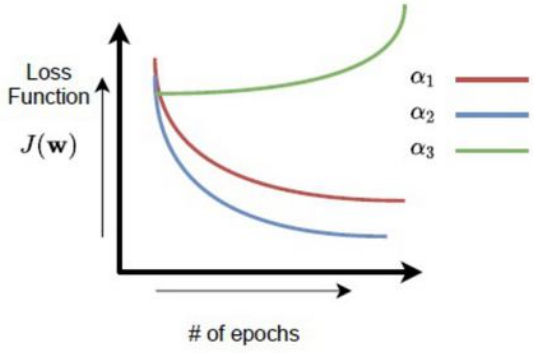
Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 187 Question Id : 640653345321 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Brijmohan trains his linear regression model with three different values of learning rate, i.e. α_1, α_2 and α_3, for the same training set. He gets learning curves as displayed in the following chart.



Which of the following correctly explains relationship among α_1, α_2 and α_3?

Options :

6406531148581. ✖ $\alpha_1 > \alpha_2 > \alpha_3$

6406531148582. ✔ $\alpha_1 < \alpha_2 < \alpha_3$

6406531148583. ✖ $\alpha_2 > \alpha_1 > \alpha_3$

6406531148584. ✖ $\alpha_3 > \alpha_2 > \alpha_1$

6406531148585. ✖ $\alpha_1 > \alpha_3 > \alpha_2$

Sub-Section Number :	6
Sub-Section Id :	64065348799
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_019 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 188 Question Id : 640653345318 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

Which of the following can NOT be a linear regression model?

Options :

6406531148571. ✖ $\mathbf{y} = \sum_{i=0}^{m} w_i x_i$

6406531148572. ✔ $\mathbf{y} = \prod_{i=0}^{m} w_i x_i$

6406531148573. ✔ $\mathbf{y} = \sum_{i=0}^{m} w_i^{x_i} x_i$

6406531148574. ✔ $\mathbf{y}^2 = \sum_{i=0}^{m} w_i^2 x_i$

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_022 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 191 Question Id : 640653345320 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

For the given dataset:

x	y
1	5
2	7
3	6

Assume the initial weight vector is $w = [1, 2]$ and learning rate $\alpha = 0.20$. Compute the weight vector after one iteration, let's call it $\mathbf{w}_1$. Compute prediction for $x = 5$ correct upto two decimal points with new weight vector (i.e. $\mathbf{w}_1$). Do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

15.4 to 15.80

✔ **Official Answer Key (Numerical Value): 15.4 to 15.80**

MLT_025 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

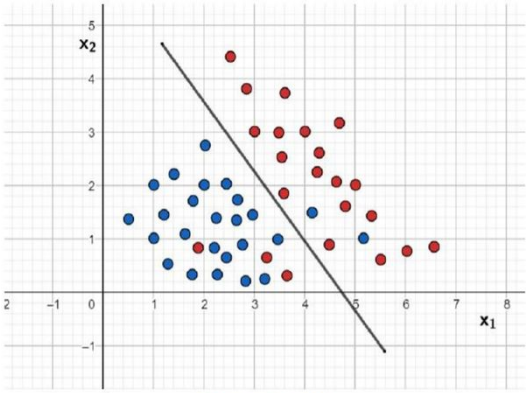
Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 194 Question Id : 640653345330 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 2

Question Label : Short Answer Question

Consider the decision boundary learnt using a perceptron algorithm as shown in the figure. Find the loss associated with this model.

Note: Since, perceptron model doesn't converge for non linearly separable data, we have terminated the learning process after a fixed numbers of iterations.



Perceptron model

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ Official Answer Key (Numerical Value): 5

MLT_026 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 195 Question Id : 640653345334 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Decision boundary for binary classification data-set is given by

$$0.3x_1 + 2.3x_2 + 1.8x_3 + 3 = 0$$

where x_1, x_2 and x_3 are features. What will be the distance of the decision boundary from the origin?

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.00 to 1.05

✔ **Official Answer Key (Numerical Value): 1.00 to 1.05**

MLT_027 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+33 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 128 Question Id : 6406531298502 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534366425. ✔ YES

6406534366426. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653206763
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLT_029 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 6406531298504 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

In a K-means algorithm with $K = 3$, the objective value $F(z_1, \ldots, z_n) = \sum_{i=1}^{n} ||x_i - \mu_{z_i}||^2$ is recorded after each iteration as follows:

Iteration 1: 800, Iteration 2: 500, Iteration 3: 400, Iteration 4: 400, Iteration 5: 400.
At which iteration is the algorithm most likely to reach convergence?

Options :

6406534366431. ✖ 2

6406534366432. ✔ 3

6406534366433. ✖ 5

6406534366434. ✖ 4

Sub-Section Number :	3
Sub-Section Id :	640653206764
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_030 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 131 Question Id : 6406531298505 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a valid kernel function such that $k(x, y) = \phi(x)^T \phi(y)$, where $\phi : \mathbb{R}^d \to \mathbb{R}^D$, where $d < D$. Define

$$k_1(x, y) = \frac{100k(x, y)}{\sqrt{k(x, x)k(y, y)}}.$$

Find $\phi_1 : \mathbb{R}^d \to \mathbb{R}^D$ such that $k_1(x, y) = \phi_1(x)^T \phi_1(y)$.

Note: $||x||_2 = \sqrt{x^T x}$.

Options :

6406534366435. ✖ $\phi_1(x) = \frac{100\phi(x)}{||\phi(x)||_2}$

6406534366436. ✖ $\phi_1(x) = \frac{\phi(x)}{10||\phi(x)||_2}$

6406534366437. ✔ $\phi_1(x) = \frac{10\phi(x)}{||\phi(x)||_2}$

6406534366438. ✖ Such a ϕ_1 does not exist, i.e., k_1 is not a valid kernel.

✔ Official Answer Key: Option (C)

MLT_031 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531298506 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Kernel PCA with a polynomial kernel is applied on the following dataset with three features:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The k-th principal component is given by $w_k = \phi(X)\alpha_k$, where ϕ is the transformation mapping corresponding to the kernel $k(x_i, x_j) = (1 + x_i^T x_j)^2$ for some $\alpha_k \in \mathbb{R}^n$. Find the centered kernel matrix K^c.

Hint: $k_{ij}^c = k_{ij} - \theta_i \mathbb{1}_j^T - \mathbb{1}_i \theta_j^T + P\mathbb{1}_i \mathbb{1}_j^T$, where $\theta_i = \frac{1}{n}\sum_{k=1}^{n} k_{ik}, \forall\, i, P = \frac{1}{n^2}\sum_{i,j} k_{ij}$.

Options :

6406534366439. ✖ $\begin{pmatrix} -1.5 & 1.5 \\ 1.5 & -1.5 \end{pmatrix}$

6406534366440. ✔ $\begin{pmatrix} 1.5 & -1.5 \\ -1.5 & 1.5 \end{pmatrix}$

6406534366441. ✖ $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

6406534366442. ✖ $\begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix}$

Sub-Section Number :	4
Sub-Section Id :	640653206765
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_035 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531298510 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset X of shape (d, n), with $d > n$. The k-th principal component of the dataset can be written as $w_k = \sum_{i=1}^{n} \alpha_i x_i$, where the vector x_i is the i-th data point. The k-th largest eigenvalue and the corresponding eigenvector of $X^T X$ are 10 and $\left[\frac{2}{\sqrt{54}} \quad \frac{3}{\sqrt{54}} \quad \frac{4}{\sqrt{54}} \quad \frac{6}{\sqrt{54}}\right]^T$, respectively. What will be the value of α_2? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.12 to 0.13

✔ **Official Answer Key (Numerical Value): 0.12 to 0.13**

MLT_036 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531298512 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample drawn from a Pareto distribution with the following probability density function:

$$f(x \mid \theta) = \theta x^{-\theta-1}, \quad x \geq 1, \ \theta > 1.$$

You are given the following sample of 5 observations:

$$X = \{2.5,\ 1.8,\ 3.2,\ 2.0,\ 2.7\}.$$

What is the maximum likelihood estimator of θ for the above sample? Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.147 to 1.150

Sub-Section Number :	7
Sub-Section Id :	640653206768
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 1.147 to 1.150**

MLT_037 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531298511 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Consider a dataset $\{x_1, x_2, \ldots, x_{50}\}$, where $x_i \in \mathbb{R}^{1000}$. After performing PCA on the dataset, suppose that the residues become zero after the fifth round. Then, how many numbers do we need to store for this dataset?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5250

Sub-Section Number :	8
Sub-Section Id :	640653206769
Question Shuffling Allowed :	No

Question Id : 6406531298513 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (139 to 141)

✔ **Official Answer Key (Numerical Value): 5250**

MLT_039 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 140 Question Id : 6406531298515 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the variance of the dataset along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

✔ **Official Answer Key (Numerical Value): 1**

MLT_040 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 141 Question Id : 6406531298516 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Find the residues after projecting the data points onto the first principal component.

Note: Here $x_i^{'}$ represents the residues of the data point x_i.

Options :

6406534366464. ✖ $x_1^{'} = \begin{bmatrix}0\\0\end{bmatrix}, \; x_2^{'} = \begin{bmatrix}0.5\\0.5\end{bmatrix}, \; x_3^{'} = \begin{bmatrix}0.5\\0.5\end{bmatrix}$

6406534366465. ✔ $x_1^{'} = \begin{bmatrix}0\\0\end{bmatrix}, \; x_2^{'} = \begin{bmatrix}0.5\\-0.5\end{bmatrix}, \; x_3^{'} = \begin{bmatrix}-0.5\\0.5\end{bmatrix}$

6406534366466. ✖ $x_1^{'} = \begin{bmatrix}0\\0\end{bmatrix}, \; x_2^{'} = \begin{bmatrix}1\\2\end{bmatrix}, \; x_3^{'} = \begin{bmatrix}2\\1\end{bmatrix}$

6406534366467. ✖ $x_1^{'} = \begin{bmatrix}0\\0\end{bmatrix}, \; x_2^{'} = \begin{bmatrix}0\\-1\end{bmatrix}, \; x_3^{'} = \begin{bmatrix}-1\\0\end{bmatrix}$

Sub-Section Number :	9
Sub-Section Id :	640653206770
Question Shuffling Allowed :	No

Question Id : 6406531298517 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (142 to 143)

✔ Official Answer Key: Option (B)

MLT_041 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 142 Question Id : 6406531298518 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Fit the linear regression model $y = w\mathbf{x}, w = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^T$ using the squared error.

Options :

6406534366468. ✖ $y = x_1 + 2x_2$

6406534366469. ✔ $y = x_1 - 2x_2$

6406534366470. ✖ $y = x_1 - x_2$

6406534366471. ✖ $y = x_1 + x_2$

✔ **Official Answer Key: Option (B)**

MLT_043 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 169 Question Id : 640653470095 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531562217. ✔ YES

6406531562218. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065367704
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLT_044 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653470096 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

Standard-PCA is performed on a centered dataset in $\mathbb{R}^3$. Two principal components are given below:

$$\frac{1}{2}\cdot\begin{bmatrix}1\\ \sqrt{2}\\ -1\end{bmatrix},\quad \frac{1}{\sqrt{2}}\cdot\begin{bmatrix}1\\ 0\\ 1\end{bmatrix}$$

Which of the following could be the third?

Options :

6406531562219. ✔ $\frac{1}{2}\cdot\begin{bmatrix}1\\ -\sqrt{2}\\ -1\end{bmatrix}$

6406531562220. ✖ $\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$

6406531562221. ✖ $\frac{1}{\sqrt{2}}\cdot\begin{bmatrix}-1\\ 0\\ 1\end{bmatrix}$

6406531562222. ✖ $\frac{1}{\sqrt{3}}\cdot\begin{bmatrix}\sqrt{2}\\ -1\\ 0\end{bmatrix}$

✔ Official Answer Key: Option (A)

MLT_045 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 640653470097 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

Consider that 1000 data points belonging to d-dimensional space have a non-linear relationship. We apply kernel PCA to reduce the dimension of the data points and take the first k principal components. Can the value of k be larger than d?

Options :

6406531562223. ✔ Yes

6406531562224. ✖ No

✔ Official Answer Key: Option (A)

MLT_046 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653470098 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

Let X be the data matrix of shape $d \times n$ with $d > n$ for a centered dataset. The eigenvector corresponding to the largest eigenvalue λ of $X^T X$ is α_1. What will be the first principal component of the dataset?

Options :

6406531562225. ✖ α_1

6406531562226. ✖ $\frac{\alpha_1}{\sqrt{\lambda}}$

6406531562227. ✖ $X\alpha_1$

6406531562228. ✔ $\frac{X\alpha_1}{\sqrt{\lambda}}$

✔ Official Answer Key: Option (D)

MLT_047 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 173 Question Id : 640653470099 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

Is the following statement true or false?

For any desired transformation $\phi(x)$, we can design a kernel function $k(x_1, x_2)$ that will evaluate $\phi(x_1)^T\phi(x_2)$.

Options :

6406531562229. ✔ TRUE

6406531562230. ✖ FALSE

✔ Official Answer Key: Option (A)

MLT_048 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

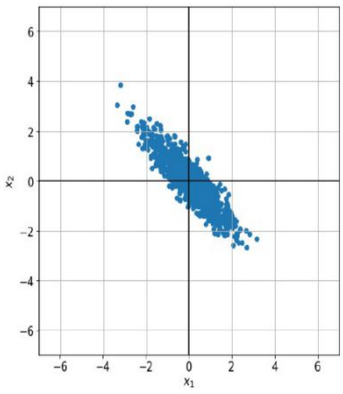
Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 174 Question Id : 640653470100 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following centered dataset in $\mathbb{R}^2$:



Which of the following could be the first principal component? Recall that the first P.C is the most important one.

Options :

6406531562231. ✖ $\begin{bmatrix}1\\0\end{bmatrix}$

6406531562232. ✖ $\begin{bmatrix}0\\1\end{bmatrix}$

6406531562233. ✖ $\frac{1}{\sqrt{2}} \cdot \begin{bmatrix}1\\1\end{bmatrix}$

6406531562234. ✔ $\frac{1}{\sqrt{2}} \cdot \begin{bmatrix}-1\\1\end{bmatrix}$

Sub-Section Number : 3

Sub-Section Id : 64065367705

✔ Official Answer Key: Option (D)

MLT_049 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 175 Question Id : 640653470101 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

In standard PCA, which of the following are correct formulations of the optimization problem? The dataset $\{\mathbf{x}_1, \cdots, \mathbf{x}_n\}$, is centered, each point lies in $\mathbb{R}^d$, and $\mathbf{C}$ is the covariance matrix. $\mathbf{w} \in \mathbb{R}^d$ is the variable that we are optimizing over.

Options :

6406531562235. ✔ $$\min_{\mathbf{w}} \quad \frac{1}{n} \cdot \sum_{i=1}^{n} ||\mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w})\mathbf{w}||^2 \quad \text{subject to} \quad ||\mathbf{w}|| = 1$$

6406531562236. ✔ $$\max_{\mathbf{w}} \quad \mathbf{w}^T \mathbf{C} \mathbf{w} \quad \text{subject to} \quad ||\mathbf{w}|| = 1$$

6406531562237. ✖ $$\max_{\mathbf{w}} \quad \frac{1}{n} \cdot \sum_{i=1}^{n} ||\mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w})\mathbf{w}||^2 \quad \text{subject to} \quad ||\mathbf{w}|| = 1$$

6406531562238. ✖ $$\min_{\mathbf{w}} \quad \mathbf{w}^T \mathbf{C} \mathbf{w} \quad \text{subject to} \quad ||\mathbf{w}|| = 1$$

✔ Official Answer Key: Option (A), Option (B)

MLT_050 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 176 Question Id : 640653470102 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

Consider a GMM with 3 components that is used to model a dataset that has 100 points. The EM algorithm is run on this dataset to estimate the parameters of the GMM. After convergence, some of the values of λ_k^i for the specific point $\mathbf{x}_{10}$ are given below:

$$\lambda_1^{10} = 0.2, \quad \lambda_2^{10} = 0.5$$

Select all true statements.

Options :

6406531562239. ✖ If a point is picked randomly from the dataset of 100 points, then there is 20% chance that it comes from the first component.

6406531562240. ✔ $\lambda_3^{10} = 0.3$

6406531562241. ✖ $\sum_{i=1}^{100} \lambda_3^i = 1$

6406531562242. ✔ There is a 50% chance that the point $\mathbf{x}_{10}$ comes from the second component.

✔ Official Answer Key: Option (B), Option (D)

MLT_051 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 177 Question Id : 640653470103 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

Consider a dataset with n points and a GMM with K components. If we fix λ and maximize for θ, what is our estimate for the mean of the k^{th} component?

Options :

6406531562243. ✔ It is the weighted mean of the n points, where the weight for point i in component k is given by λ_k^i

6406531562244. ✖ It is the mean of the n points.

6406531562245. ✖ $$\hat{\mu}_k = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$$

6406531562246. ✔ $$\hat{\mu}_k = \frac{\sum_{i=1}^{n} \lambda_k^i x_i}{\sum_{i=1}^{n} \lambda_k^i}$$

✔ Official Answer Key: Option (A), Option (D)

MLT_053 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 179 Question Id : 640653470105 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

Consider a dataset of 20 points where the i^{th} data-point is given by:

$$\mathbf{x}_i = a_i \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} + b_i \cdot \begin{bmatrix} -1 \\ 1 \\ 3 \\ 0 \end{bmatrix}$$

where, a_i and b_i are real numbers such that $\sum_{i=1}^{20} a_i = \sum_{i=1}^{20} b_i = 0$. Standard PCA is performed on this dataset. If the top two principal components are retained and used to reconstruct the dataset, what is the reconstruction error?

Hint: Think about what happens in $\mathbb{R}^2$ or $\mathbb{R}^3$ for a similar situation and extend this idea to $\mathbb{R}^4$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

✔ **Official Answer Key (Numerical Value): 0**

MLT_054 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 180 Question Id : 640653470106 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

Consider the following prior for the parameter p of a Bernoulli distribution:

$$p \sim \text{Beta}(3, 2)$$

The dataset observed is as follows:

$$\{1, \quad 0, \quad 1, \quad 0, \quad 1, \quad 1, \quad 1, \quad 0, \quad 0, \quad 0, \quad 1, \quad 1, \quad 1\}$$

What is $\hat{p}$, a point estimate for the parameter of the Bernoulli distribution, if we use the expectation of the posterior as the method of estimation? Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.60 to 0.62

✔ **Official Answer Key (Numerical Value): 0.60 to 0.62**

MLT_056 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 182 Question Id : 640653470109 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2.5

Question Label : Short Answer Question

Each data-point in D_2 belongs to $\mathbb{R}^k$. What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ **Official Answer Key (Numerical Value): 5**

MLT_057 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 183 Question Id : 640653470110 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2.5

Question Label : Short Answer Question

If each real number occupies unit storage space in the memory, compute:

$$\frac{\text{size}(D_1)}{\text{size}(D_2)}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Sub-Section Number :	6
Sub-Section Id :	64065367708
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653470111 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (184 to 185)

Question Label : Comprehension

Assume that you have a dataset of five points $\{x_1, x_2, x_3, x_4, x_5\}$, all of which are non-negative. You hypothesise that the data points are iid random variables with the following density:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key (Numerical Value): 20

MLT_058 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 184 Question Id : 640653470112 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 5

Question Label : Multiple Choice Question

What is the log-likelihood of this dataset under this distribution? ln represents the natural logarithm or $\log_e$.

Options :

6406531562258. ✖ $\prod_{i=1}^{5} \lambda e^{-\lambda x_i}$

6406531562259. ✖ $\sum_{i=1}^{5} \lambda e^{-\lambda x_i}$

6406531562260. ✔ $\sum_{i=1}^{5} [\ln(\lambda) - \lambda x_i]$

6406531562261. ✖ $\prod_{i=1}^{5} [\ln(\lambda) - \lambda x_i]$

6406531562262. ✖ $\sum_{i=1}^{5} [\ln(\lambda) - \lambda \ln(x_i)]$

✔ Official Answer Key: Option (C)

MLT_059 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 185 Question Id : 640653470113 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5$

What is the maximum likelihood estimate for λ?
Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.32 to 0.34

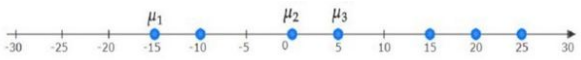
Sub-Section Number :	7
Sub-Section Id :	64065367709
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653470114 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (186 to 190)

Question Label : Comprehension

Consider the following one-dimensional dataset of seven points that are distributed as follows:



k-means algorithm with $k = 3$ was run on the given data points. μ_1, μ_2, and μ_3 are the initial cluster means.

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key (Numerical Value): 0.32 to 0.34

MLT_060 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 186 Question Id : 640653470115 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

How many points belong to cluster 3 (mean μ_3) for the initial clusters?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ **Official Answer Key (Numerical Value): 4**

MLT_061 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 187 Question Id : 640653470116 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What will be the mean μ_3 after 1^{st} iteration? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

16.23 to 16.27

✔ **Official Answer Key (Numerical Value): 16.23 to 16.27**

MLT_062 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 188 Question Id : 640653470117 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

How many points belong to cluster 2 (mean μ_2) after the 1^{st} iteration?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

✔ **Official Answer Key (Numerical Value): 2**

MLT_063 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 189 Question Id : 640653470118 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

How many iterations will it take for the algorithm to converge with the given initial clusters?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

✔ **Official Answer Key (Numerical Value): 2**

MLT_067 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+6 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 179 Question Id : 640653737457 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532468141. ✔ YES

6406532468142. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653107658
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLT_068 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 180 Question Id : 640653737458 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose a dataset lies in $\mathbb{R}^4$ and undergoes Principal Component Analysis (PCA) after being centered. The resulting first and second principal components are given by:

$$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Which of the following could be the third principal component?

Options :

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

6406532468143. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

6406532468144. ✖

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$

6406532468145. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

6406532468146. ✔

✔ Official Answer Key: Option (D)

MLT_069 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 181 Question Id : 640653737462 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel. Is $\sin(x_1)\cos(x_2)k(x_1, x_2)$ a valid kernel? Here, $x_1, x_2 \in \mathbb{R}$?

Options :

6406532468149. ✖ Yes

6406532468150. ✔ No

✔ Official Answer Key: Option (B)

MLT_070 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 182 Question Id : 640653737463 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406532468151. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406532468152. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406532468153. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406532468154. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

✔ **Official Answer Key: Option (A)**

MLT_072 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 184 Question Id : 640653737460 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

What is the value of a? Enter your answer correctly to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.10 to 1.20

✔ **Official Answer Key (Numerical Value): 1.10 to 1.20**

MLT_073 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 185 Question Id : 640653737461 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

What is the value of *b*? Enter your answer correctly to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.65 to 0.75

Question Id : 640653737466 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (186 to 187)

✔ **Official Answer Key (Numerical Value): 0.65 to 0.75**

MLT_077 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 189 Question Id : 640653737472 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5, \quad x_6 = 6$

What is the maximum likelihood estimate for λ? Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.25 to 0.35

Sub-Section Number :	5
Sub-Section Id :	640653107661
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.25 to 0.35**

MLT_079 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 191 Question Id : 640653737473 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

For the given Beta distribution, choose the correct option.

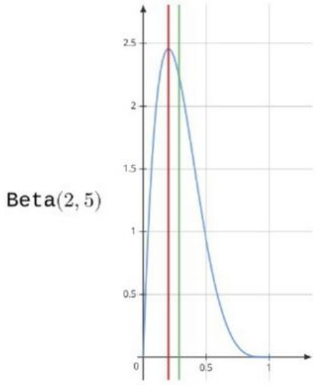


Figure 1: Beta(2,5)

Options :

6406532468175. ✖ Red line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532468176. ✔ Green line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532468177. ✔ Red line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

6406532468178. ✖ Green line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

Sub-Section Number :	6
Sub-Section Id :	640653107662
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (C)

MLT_081 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 180 Question Id : 640653737744 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose a dataset lies in $\mathbb{R}^4$ and undergoes Principal Component Analysis (PCA) after being centered. The resulting first and second principal components are given by:

$$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Which of the following could be the third principal component?

Options :

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

6406532468967. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

6406532468968. ✖

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$

6406532468969. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

6406532468970. ✔

✔ Official Answer Key: Option (D)

MLT_082 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 181 Question Id : 640653737748 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel. Is $\sin(x_1)\cos(x_2)k(x_1, x_2)$ a valid kernel? Here, $x_1, x_2 \in \mathbb{R}$?

Options :

6406532468973. ✖ Yes

6406532468974. ✔ No

✔ Official Answer Key: Option (B)

MLT_083 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 182 Question Id : 640653737749 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406532468975. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406532468976. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406532468977. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406532468978. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

✔ **Official Answer Key: Option (A)**

MLT_084 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 189 Question Id : 640653737758 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5, \quad x_6 = 6$

What is the maximum likelihood estimate for λ? Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.25 to 0.35

Sub-Section Number :	5
Sub-Section Id :	640653107774
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.25 to 0.35**

MLT_086 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 191 Question Id : 640653737759 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

For the given Beta distribution, choose the correct option.

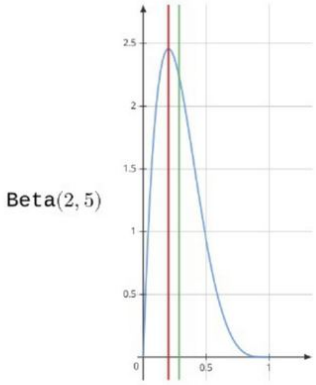


Figure 1: Beta(2,5)

Options :

6406532468999. ✖ Red line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532469000. ✔ Green line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532469001. ✔ Red line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

6406532469002. ✖ Green line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

Sub-Section Number :	6
Sub-Section Id :	640653107775
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (C)

MLT_087 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 180 Question Id : 640653738030 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose a dataset lies in $\mathbb{R}^4$ and undergoes Principal Component Analysis (PCA) after being centered. The resulting first and second principal components are given by:

$$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Which of the following could be the third principal component?

Options :

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

6406532469791. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

6406532469792. ✖

$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$

6406532469793. ✖

$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

6406532469794. ✔

✔ Official Answer Key: Option (D)

MLT_088 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 181 Question Id : 640653738034 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel. Is $\sin(x_1)\cos(x_2)k(x_1, x_2)$ a valid kernel? Here, $x_1, x_2 \in \mathbb{R}$?

Options :

6406532469797. ✖ Yes

6406532469798. ✔ No

✔ Official Answer Key: Option (B)

MLT_089 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 182 Question Id : 640653738035 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406532469799. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406532469800. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406532469801. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406532469802. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

✔ Official Answer Key: Option (A)

MLT_090 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 189 Question Id : 640653738044 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5, \quad x_6 = 6$

What is the maximum likelihood estimate for λ? Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.25 to 0.35

Sub-Section Number :	5
Sub-Section Id :	640653107887
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.25 to 0.35**

MLT_092 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 191 Question Id : 640653738045 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

For the given Beta distribution, choose the correct option.

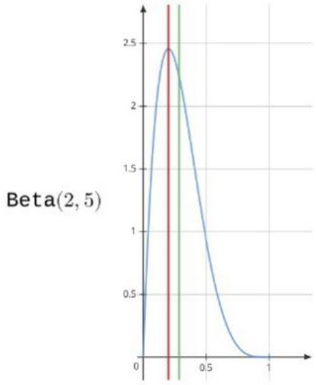


Figure 1: Beta(2,5)

Options :

6406532469823. ✖ Red line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532469824. ✔ Green line denotes the mean for the given Beta distribution, i.e. mean $= \frac{2}{7}$

6406532469825. ✔ Red line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

6406532469826. ✖ Green line denotes the mode for the given Beta distribution, i.e. mode $= \frac{1}{5}$

Sub-Section Number :	6
Sub-Section Id :	640653107888
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (C)

MLT_093 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531113827 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

For the Lloyd's algorithm, what would be the correct relationship among the following three quantities?

(1) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t+1}}||^2$,

(2) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t}}||^2$,

(3) $\sum_{i=1}^{n} ||x_i - \mu^{t+1}_{z_i^{t+1}}||^2$

where $\mu^{t}_{z_i^{t}}$ and $\mu^{t+1}_{z_i^{t+1}}$ refer to means of cluster z_i in iterations, t and $t+1$ respectively, and $\mu^{t}_{z_i^{t+1}}$ refer to the mean of the cluster z_i where x_i is going to move in the next iteration, i.e., $(t+1)$-th iteration.

Options :

6406533774393. ✖ (1) > (2) > (3)

6406533774394. ✖ (1) < (2) < (3)

6406533774395. ✔ (3) < (1) < (2)

6406533774396. ✖ (2) < (3) < (1)

Sub-Section Number : 3

Sub-Section Id : 640653168532

Question Shuffling Allowed : Yes

✔ Official Answer Key: Option (C)

MLT_094 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 133 Question Id : 6406531113828 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

K-means clustering is run on a dataset of 100 points with $K = 3$. If you know that the points $\begin{bmatrix}0\\1\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are a part of the same cluster $C1$, then which of the following points will definitely lie in $C1$?

Options :

6406533774397. ✖ $\begin{bmatrix}-1\\1\end{bmatrix}$

6406533774398. ✖ $\begin{bmatrix}1\\2\end{bmatrix}$

6406533774399. ✔ $\begin{bmatrix}0.2\\1\end{bmatrix}$

6406533774400. ✖ $\begin{bmatrix}0\\0\end{bmatrix}$

✔ Official Answer Key: Option (C)

MLT_095 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 134 Question Id : 6406531113829 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $x_1, x_2, \ldots, x_n$ be d-dimensional data points $(d > n)$ and X be the matrix of shape $d \times n$ containing the data points. The k^{th} largest eigenvalue and corresponding unit eigenvector of $X^T X$ is λ and α_k, respectively. What will be the projection of x_i on the k^{th} principal component?

Options :

6406533774401. ✖ $x_i^T \alpha_k$

6406533774402. ✖ $\frac{x_i^T \alpha_k}{\sqrt{n\lambda}}$

6406533774403. ✔ $\frac{x_i^T X \alpha_k}{\sqrt{\lambda}}$

6406533774404. ✖ $x_i^T X \alpha_k$

Sub-Section Number :	4
Sub-Section Id :	640653168533
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C)

MLT_096 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 135 Question Id : 6406531113830 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following matrices are appropriate matrix $K = X^T X$ for some data matrix X? Choose the most appropriate answer.

Options :

6406533774405. ✔ $K = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 5 & 3 \\ 2 & 3 & 6 \end{bmatrix}$

6406533774406. ✖ $K = \begin{bmatrix} -2 & -1 & 0 \\ -1 & -3 & -1 \\ 0 & -1 & -1 \end{bmatrix}$

6406533774407. ✖ $K = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 5 & -2 \\ 1 & -2 & 2 \end{bmatrix}$

6406533774408. ✔ $K = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 5 \end{bmatrix}$

Sub-Section Number :	5
Sub-Section Id :	640653168534
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_097 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531113831 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are true?
Options :

6406533774409. ✔ $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, $k(x_1, x_2) = (x_1^T x_2)^3$ is a valid kernel.

6406533774410. ✔ $k : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $k(x_1, x_2) = e^{-\|x_1 - x_2\|^2}$ is a valid kernel.

6406533774411. ✔ The trace of XX^T is equal to the trace of $X^T X$.

6406533774412. ✖ Eigenvectors of XX^T are the same as eigenvectors of $X^T X$.

Sub-Section Number :	6
Sub-Section Id :	640653168535
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_098 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531113832 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let X be a data matrix of shape (d, n) for a centered dataset. The first principal component of the dataset is given as

$$\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]^T.$$

What will be the scalar proxy of the point $\begin{bmatrix}3\\4\end{bmatrix}$ onto the first principal component?

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.92 to 4.98

✔ **Official Answer Key (Numerical Value): 4.92 to 4.98**

MLT_099 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531113833 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset that has 1000 samples, where each sample belongs to $\mathbb{R}^{20}$. PCA is run on this dataset and the top 5 principal components are retained, the rest being discarded. If it takes one unit of memory to store a real number, find the percentage decrease in storage space of the dataset by moving to its compressed representation. Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

74 to 75

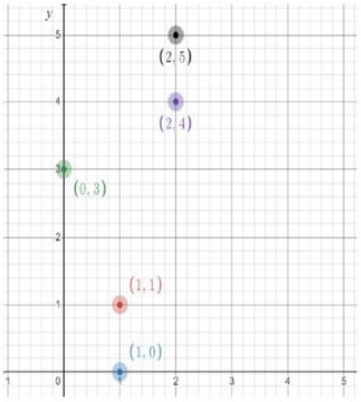
✔ **Official Answer Key (Numerical Value): 74 to 75**

MLT_100 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 139 Question Id : 6406531113834 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

K-means++ algorithm was run on the following dataset with $K = 2$. If we



use the point $(1, 1)$ as the first cluster mean, then find the probability of the point that has the least chance of being chosen as the second cluster mean. Enter the answer correct to two decimal places. Use Euclidean distance to calculate the distance.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.07 to 0.11

✔ Official Answer Key (Numerical Value): 0.07 to 0.11

MLT_102 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 141 Question Id : 6406531113836 Question Type : SA
Correct Marks : 4

Question Label : Short Answer Question

Suppose we fit a Gaussian Mixture Model with $K = 2$ for the following dataset:

$$D = \{x_1 = 1, x_2 = 3, x_3 = 2, x_4 = 5\}.$$

At the beginning of the t-th time step of the EM algorithm, we have $\theta^{(t)}$ as follows:

$$\pi_1 = 0.5, \quad \pi_2 = 0.5$$

$$\mu_1 = 1, \quad \mu_2 = 3$$

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

What is the probability that x_1 belongs to the first component? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.86 to 0.90

Sub-Section Number :	8
Sub-Section Id :	640653168537
Question Shuffling Allowed :	No

Question Id : 6406531113837 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

Question Label : Comprehension

Consider the following dataset consisting of four points, all of which are collinear:

$$S = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \end{bmatrix} \right\}$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 0.86 to 0.90**

MLT_103 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 142 Question Id : 6406531113838 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Among the vectors given below, choose a representative that has unit length.

Options :

6406533774418. ✖ $\begin{bmatrix} 1 \\ \frac{3}{2} \\ \frac{1}{3} \end{bmatrix}$

6406533774419. ✔ $\begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$

6406533774420. ✖ $\begin{bmatrix} \frac{3}{\sqrt{5}} \\ \frac{6}{\sqrt{5}} \end{bmatrix}$

6406533774421. ✖ $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$

✔ Official Answer Key: Option (B)

MLT_104 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 143 Question Id : 6406531113839 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If standard PCA is performed on this dataset, what is the variance along the first principal component? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

43 to 44

Sub-Section Number :	9
Sub-Section Id :	640653168538
Question Shuffling Allowed :	No

Question Id : 6406531113840 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (144 to 145)

Question Label : Comprehension

Suppose that 100 items are sampled from a manufacturing process and 10 are found to be defective. Let p denote the proportion of defective items. Assume that the prior distribution for p is Beta(1, 1).

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 43 to 44**

MLT_107 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531114454 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

For the Lloyd's algorithm, what would be the correct relationship among the following three quantities?

(1) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t+1}}||^2$,

(2) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t}}||^2$,

(3) $\sum_{i=1}^{n} ||x_i - \mu^{t+1}_{z_i^{t+1}}||^2$

where $\mu^{t}_{z_i^{t}}$ and $\mu^{t+1}_{z_i^{t+1}}$ refer to means of cluster z_i in iterations, t and $t+1$ respectively, and $\mu^{t}_{z_i^{t+1}}$ refer to the mean of the cluster z_i where x_i is going to move in the next iteration, i.e., $(t+1)$-th iteration.

Options :

6406533776206. ✖ (1) > (2) > (3)

6406533776207. ✖ (1) < (2) < (3)

6406533776208. ✔ (3) < (1) < (2)

6406533776209. ✖ (2) < (3) < (1)

Sub-Section Number : 3

Sub-Section Id : 640653168736

Question Shuffling Allowed : Yes

✔ Official Answer Key: Option (C)

MLT_108 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 133 Question Id : 6406531114455 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

K-means clustering is run on a dataset of 100 points with $K = 3$. If you know that the points $\begin{bmatrix}0\\1\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are a part of the same cluster $C1$, then which of the following points will definitely lie in $C1$?

Options :

6406533776210. ✖ $\begin{bmatrix}-1\\1\end{bmatrix}$

6406533776211. ✖ $\begin{bmatrix}1\\2\end{bmatrix}$

6406533776212. ✔ $\begin{bmatrix}0.2\\1\end{bmatrix}$

6406533776213. ✖ $\begin{bmatrix}0\\0\end{bmatrix}$

✔ Official Answer Key: Option (C)

MLT_109 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 134 Question Id : 6406531114456 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $x_1, x_2, \ldots, x_n$ be d-dimensional data points $(d > n)$ and X be the matrix of shape $d \times n$ containing the data points. The k^{th} largest eigenvalue and corresponding unit eigenvector of $X^T X$ is λ and α_k, respectively. What will be the projection of x_i on the k^{th} principal component?

Options :

6406533776214. ✖ $x_i^T \alpha_k$

6406533776215. ✖ $\frac{x_i^T \alpha_k}{\sqrt{n\lambda}}$

6406533776216. ✔ $\frac{x_i^T X \alpha_k}{\sqrt{\lambda}}$

6406533776217. ✖ $x_i^T X \alpha_k$

Sub-Section Number :	4
Sub-Section Id :	640653168737
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C)

MLT_110 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 135 Question Id : 6406531114457 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following matrices are appropriate matrix $K = X^T X$ for some data matrix X? Choose the most appropriate answer.

Options :

6406533776218. ✔ $K = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 5 & 3 \\ 2 & 3 & 6 \end{bmatrix}$

6406533776219. ✖ $K = \begin{bmatrix} -2 & -1 & 0 \\ -1 & -3 & -1 \\ 0 & -1 & -1 \end{bmatrix}$

6406533776220. ✖ $K = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 5 & -2 \\ 1 & -2 & 2 \end{bmatrix}$

6406533776221. ✔ $K = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 5 \end{bmatrix}$

Sub-Section Number :	5
Sub-Section Id :	640653168738
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_111 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531114458 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are true?
Options :

6406533776222. ✔ $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, $k(x_1, x_2) = (x_1^T x_2)^3$ is a valid kernel.

6406533776223. ✔ $k : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $k(x_1, x_2) = e^{-\|x_1 - x_2\|^2}$ is a valid kernel.

6406533776224. ✔ The trace of XX^T is equal to the trace of $X^T X$.

6406533776225. ✖ Eigenvectors of XX^T are the same as eigenvectors of $X^T X$.

Sub-Section Number :	6
Sub-Section Id :	640653168739
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_112 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531114459 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let X be a data matrix of shape (d, n) for a centered dataset. The first principal component of the dataset is given as

$$\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]^T.$$

What will be the scalar proxy of the point $\begin{bmatrix}3\\4\end{bmatrix}$ onto the first principal component?

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.92 to 4.98

✔ **Official Answer Key (Numerical Value): 4.92 to 4.98**

MLT_113 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531114460 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset that has 1000 samples, where each sample belongs to $\mathbb{R}^{20}$. PCA is run on this dataset and the top 5 principal components are retained, the rest being discarded. If it takes one unit of memory to store a real number, find the percentage decrease in storage space of the dataset by moving to its compressed representation. Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

74 to 75

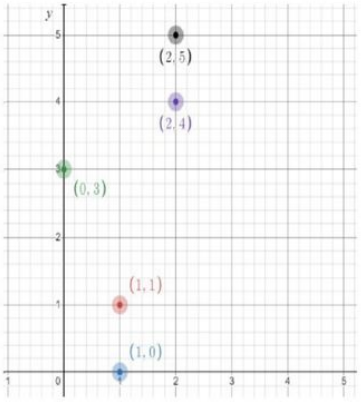
✔ **Official Answer Key (Numerical Value): 74 to 75**

MLT_114 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 139 Question Id : 6406531114461 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

K-means++ algorithm was run on the following dataset with $K = 2$. If we



use the point $(1, 1)$ as the first cluster mean, then find the probability of the point that has the least chance of being chosen as the second cluster mean. Enter the answer correct to two decimal places. Use Euclidean distance to calculate the distance.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.07 to 0.11

✔ Official Answer Key (Numerical Value): 0.07 to 0.11

MLT_115 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 141 Question Id : 6406531114463 Question Type : SA
Correct Marks : 4

Question Label : Short Answer Question

Suppose we fit a Gaussian Mixture Model with $K = 2$ for the following dataset:

$$D = \{x_1 = 1, x_2 = 3, x_3 = 2, x_4 = 5\}.$$

At the beginning of the t-th time step of the EM algorithm, we have $\theta^{(t)}$ as follows:

$$\pi_1 = 0.5, \quad \pi_2 = 0.5$$

$$\mu_1 = 1, \quad \mu_2 = 3$$

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

What is the probability that x_1 belongs to the first component? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.86 to 0.90

Sub-Section Number :	8
Sub-Section Id :	640653168741
Question Shuffling Allowed :	No

Question Id : 6406531114464 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

Question Label : Comprehension

Consider the following dataset consisting of four points, all of which are collinear:

$$S = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \end{bmatrix} \right\}$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 0.86 to 0.90**

MLT_116 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 143 Question Id : 6406531114466 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If standard PCA is performed on this dataset, what is the variance along the first principal component? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

43 to 44

Sub-Section Number :	9
Sub-Section Id :	640653168742
Question Shuffling Allowed :	No

Question Id : 6406531114467 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (144 to 145)

Question Label : Comprehension

Suppose that 100 items are sampled from a manufacturing process and 10 are found to be defective. Let p denote the proportion of defective items. Assume that the prior distribution for p is Beta(1, 1).

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 43 to 44**

MLT_117 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531114167 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

For the Lloyd's algorithm, what would be the correct relationship among the following three quantities?

(1) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t+1}}||^2$,

(2) $\sum_{i=1}^{n} ||x_i - \mu^{t}_{z_i^{t}}||^2$,

(3) $\sum_{i=1}^{n} ||x_i - \mu^{t+1}_{z_i^{t+1}}||^2$

where $\mu^{t}_{z_i^{t}}$ and $\mu^{t+1}_{z_i^{t+1}}$ refer to means of cluster z_i in iterations, t and $t+1$ respectively, and $\mu^{t}_{z_i^{t+1}}$ refer to the mean of the cluster z_i where x_i is going to move in the next iteration, i.e., $(t+1)$-th iteration.

Options :

6406533775377. ✖ (1) > (2) > (3)

6406533775378. ✖ (1) < (2) < (3)

6406533775379. ✔ (3) < (1) < (2)

6406533775380. ✖ (2) < (3) < (1)

Sub-Section Number : 3

Sub-Section Id : 640653168637

Question Shuffling Allowed : Yes

✔ Official Answer Key: Option (C)

MLT_118 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 133 Question Id : 6406531114168 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

K-means clustering is run on a dataset of 100 points with $K = 3$. If you know that the points $\begin{bmatrix}0\\1\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are a part of the same cluster $C1$, then which of the following points will definitely lie in $C1$?

Options :

6406533775381. ✖ $\begin{bmatrix}-1\\1\end{bmatrix}$

6406533775382. ✖ $\begin{bmatrix}1\\2\end{bmatrix}$

6406533775383. ✔ $\begin{bmatrix}0.2\\1\end{bmatrix}$

6406533775384. ✖ $\begin{bmatrix}0\\0\end{bmatrix}$

✔ Official Answer Key: Option (C)

MLT_119 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 134 Question Id : 6406531114169 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $x_1, x_2, \ldots, x_n$ be d-dimensional data points $(d > n)$ and X be the matrix of shape $d \times n$ containing the data points. The k^{th} largest eigenvalue and corresponding unit eigenvector of $X^T X$ is λ and α_k, respectively. What will be the projection of x_i on the k^{th} principal component?

Options :

6406533775385. ✖ $x_i^T \alpha_k$

6406533775386. ✖ $\frac{x_i^T \alpha_k}{\sqrt{n\lambda}}$

6406533775387. ✔ $\frac{x_i^T X \alpha_k}{\sqrt{\lambda}}$

6406533775388. ✖ $x_i^T X \alpha_k$

Sub-Section Number :	4
Sub-Section Id :	640653168638
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C)

MLT_120 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 135 Question Id : 6406531114170 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following matrices are appropriate matrix $K = X^T X$ for some data matrix X? Choose the most appropriate answer.

Options :

6406533775389. ✔ $K = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 5 & 3 \\ 2 & 3 & 6 \end{bmatrix}$

6406533775390. ✖ $K = \begin{bmatrix} -2 & -1 & 0 \\ -1 & -3 & -1 \\ 0 & -1 & -1 \end{bmatrix}$

6406533775391. ✖ $K = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 5 & -2 \\ 1 & -2 & 2 \end{bmatrix}$

6406533775392. ✔ $K = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 5 \end{bmatrix}$

Sub-Section Number :	5
Sub-Section Id :	640653168639
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_121 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531114171 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are true?
Options :

6406533775393. ✔ $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, $k(x_1, x_2) = (x_1^T x_2)^3$ is a valid kernel.

6406533775394. ✔ $k : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $k(x_1, x_2) = e^{-\|x_1 - x_2\|^2}$ is a valid kernel.

6406533775395. ✔ The trace of XX^T is equal to the trace of $X^T X$.

6406533775396. ✖ Eigenvectors of XX^T are the same as eigenvectors of $X^T X$.

Sub-Section Number :	6
Sub-Section Id :	640653168640
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_122 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531114172 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let X be a data matrix of shape (d, n) for a centered dataset. The first principal component of the dataset is given as

$$\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]^T.$$

What will be the scalar proxy of the point $\begin{bmatrix}3\\4\end{bmatrix}$ onto the first principal component?

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.92 to 4.98

✔ **Official Answer Key (Numerical Value): 4.92 to 4.98**

MLT_123 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531114173 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset that has 1000 samples, where each sample belongs to $\mathbb{R}^{20}$. PCA is run on this dataset and the top 5 principal components are retained, the rest being discarded. If it takes one unit of memory to store a real number, find the percentage decrease in storage space of the dataset by moving to its compressed representation. Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

74 to 75

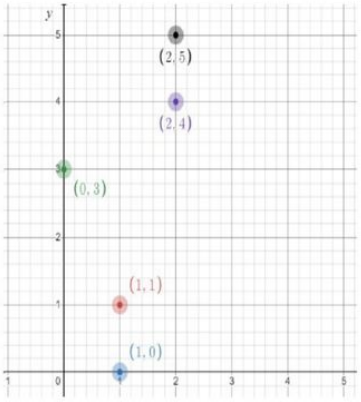
✔ **Official Answer Key (Numerical Value): 74 to 75**

MLT_124 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 139 Question Id : 6406531114174 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

K-means++ algorithm was run on the following dataset with $K = 2$. If we



use the point $(1, 1)$ as the first cluster mean, then find the probability of the point that has the least chance of being chosen as the second cluster mean. Enter the answer correct to two decimal places. Use Euclidean distance to calculate the distance.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.07 to 0.11

✔ Official Answer Key (Numerical Value): 0.07 to 0.11

MLT_125 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 141 Question Id : 6406531114176 Question Type : SA
Correct Marks : 4

Question Label : Short Answer Question

Suppose we fit a Gaussian Mixture Model with $K = 2$ for the following dataset:

$$D = \{x_1 = 1, x_2 = 3, x_3 = 2, x_4 = 5\}.$$

At the beginning of the t-th time step of the EM algorithm, we have $\theta^{(t)}$ as follows:

$$\pi_1 = 0.5, \quad \pi_2 = 0.5$$

$$\mu_1 = 1, \quad \mu_2 = 3$$

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

What is the probability that x_1 belongs to the first component? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.86 to 0.90

Sub-Section Number :	8
Sub-Section Id :	640653168642
Question Shuffling Allowed :	No

Question Id : 6406531114177 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

Question Label : Comprehension

Consider the following dataset consisting of four points, all of which are collinear:

$$S = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \end{bmatrix} \right\}$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 0.86 to 0.90**

MLT_126 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 143 Question Id : 6406531114179 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If standard PCA is performed on this dataset, what is the variance along the first principal component? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

43 to 44

Sub-Section Number :	9
Sub-Section Id :	640653168643
Question Shuffling Allowed :	No

Question Id : 6406531114180 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (144 to 145)

Question Label : Comprehension

Suppose that 100 items are sampled from a manufacturing process and 10 are found to be defective. Let p denote the proportion of defective items. Assume that the prior distribution for p is Beta(1, 1).

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 43 to 44**

MLT_128 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 3 Question Id : 6406531951435

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a centered dataset D in $\mathbb{R}^4$ such that every data-point (x_1, x_2, x_3, x_4) in D satisfies the equation $2x_1 + x_2 - x_3 + 2x_4 = 0$. Standard PCA is performed on the dataset. Which of the following could be the fourth principal component?

Options :

6406536344675. ✔ $\frac{1}{\sqrt{10}}(2, 1, -1, 2)$

6406536344676. ✔ $\frac{1}{\sqrt{10}}(-2, -1, 1, -2)$

6406536344677. ✖ $\frac{1}{\sqrt{6}}(1, 0, 2, 1)$

6406536344678. ✖ $\frac{1}{3}(2, 0, 1, 2)$

Sub-Section Number :	4
Sub-Section Id :	640653360304
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_133 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Label : Comprehension

Question Number : 8 Question Id : 6406531951441

Correct Marks : 2

Question Label : Short Answer Question

Let $v \in \mathbb{R}^3$ be some unit vector. Find the variance of the dataset along the direction v.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Sub-Section Number :	6
Sub-Section Id :	640653360306
Question Shuffling Allowed :	No

Question Id : 6406531951442 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (9 to 10)

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_134 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Label : Comprehension

$\mathbb{D}^3$

Question Number : 9 Question Id : 6406531951443

Correct Marks : 2

Question Label : Short Answer Question

Find a. Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.071 to 2.171

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_135 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Label : Comprehension

$\mathbb{D}^3$

Question Number : 10 Question Id : 6406531951444

Correct Marks : 2

Question Label : Short Answer Question

Find b. Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.657 to 0.757

Question Id : 6406531951445 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (11 to 12)

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_142 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 169 Question Id : 640653577922 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel function:

$$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$$

$$k([x_1, x_2]^T, [y_1, y_2]^T) = 1 + x_1y_1 + x_2y_2 + x_1^2y_1^2x_2y_2 + x_1y_1x_2^2y_2^2$$

Find the appropriate transformation mapping ϕ for the given kernel.

Options :

6406531929891. ✖ $\phi([x_1, x_2]^T) = [1,\ x_1x_2,\ x_1^2x_2^2]$

6406531929892. ✖ $\phi([x_1, x_2]^T) = [1,\ x_1,\ x_2,\ x_1^2x_2^2]$

6406531929893. ✔ $\phi([x_1, x_2]^T) = [1,\ x_1,\ x_2,\ x_1^2x_2,\ x_1x_2^2]$

6406531929894. ✖ $\phi([x_1, x_2]^T) = [1,\ x_1,\ x_2,\ x_1x_2,\ x_1^2x_2^2]$

✔ **Official Answer Key: Option (C)**

MLT_143 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653577923 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Hard K-means clustering problem with four points: $P(1,1), Q(2,1), R(4,3)$, and $S(5,4)$. Consider the number of clusters to be $k = 2$ and the initial centroids to be $C_1 = (0,0)$ and $C_2 = (4,4)$. After how many iterations will the algorithm terminate and what will be the final centroids? use euclidean distance metric to calculate distance.

Options :

6406531929895. ✖ $2, (1,1), (4,4)$

6406531929896. ✖ $3, (1,1), (4,4)$

6406531929897. ✔ $2, (1.5,1), (4.5,3.5)$

6406531929898. ✖ $4, (1.5,1), (4.5,3.5)$

Sub-Section Number :	3
Sub-Section Id :	64065382613
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C)

MLT_144 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 640653577924 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select all true statements regarding standard PCA applied on a centered dataset.

Options :

6406531929899. ✔ The variance of the dataset along the first principal component is maximum.

6406531929900. ✖ The variance of the dataset along the first principal component is minimum.

6406531929901. ✔ The first principal component is the unit norm eigenvector corresponding to the largest eigenvalue of the covariance matrix.

6406531929902. ✖ The first principal component is the unit norm eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

✔ Official Answer Key: Option (A), Option (C)

MLT_145 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653577925 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset of 2000 points all of which lie in $\mathbb{R}^{30}$. If all these points lie in four dimensional low subspace then after applying the PCA algorithm which of the following statements is/are true?

Options :

6406531929903. ✖ The covariance matrix corresponding to this dataset has 30 non-zero eigenvalues.

6406531929904. ✔ The covariance matrix corresponding to this dataset has only four non-zero eigenvalues.

6406531929905. ✔ The residues will become zero after four rounds

6406531929906. ✔ By utilizing four principal component vectors and their corresponding coefficients, we have the ability to reconstruct the complete dataset using representatives.

Sub-Section Number :	4
Sub-Section Id :	64065382614
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_146 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 173 Question Id : 640653577926 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements about kernel PCA is/are always true?

Options :

6406531929907. ✖ The number of principal components corresponding to nonzero eigenvalues obtained using kernel PCA on d dimensional dataset cannot be more than d.

6406531929908. ✔ For every transformation mapping $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^D$, there exists a valid kernel function.

6406531929909. ✔ For two valid kernels $k_1, k_2 : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, $ak_1 + bk_2$ is a valid kernel function. Here both a and b are positive real numbers.

6406531929910. ✖ The output of a valid kernel function can never be a negative real number.

✔ Official Answer Key: Option (B), Option (C)

MLT_147 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

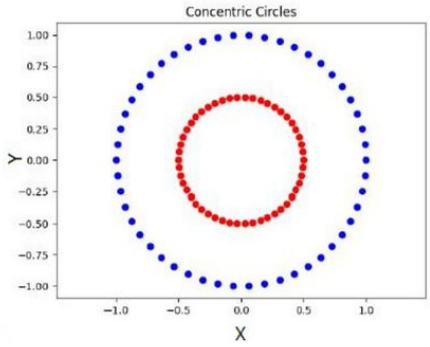
Question Number : 174 Question Id : 640653577928 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

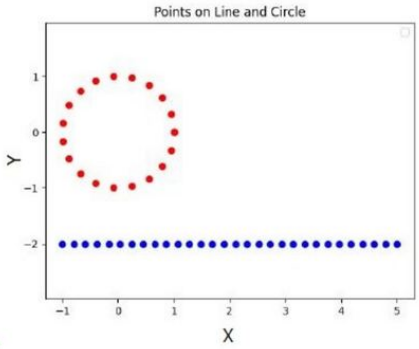
Question Label : Multiple Select Question

Which of the following clusters cannot be obtained as final clusters using Lloyd's algorithm? Assume that Lloyd's algorithm was run on these datasets with $k = 2$.

Options :



6406531929915. ✔



6406531929916. ✔

6406531929917. ✖

✔ Official Answer Key: Option (A), Option (B)

MLT_148 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 175 Question Id : 640653577927 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is/are true?

Options :

6406531929911. ✔ $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}, k(x_1, x_2) = (x_1^T x_2)^3$ is a valid kernel.

6406531929912. ✖ $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}, k(x_1, x_2) = -(x_1^T x_2)^3$ is a valid kernel.

6406531929913. ✔ nonzero Eigenvalues of XX^T and $X^T X$ are the same.

6406531929914. ✖ Eigenvectors of XX^T are same as eigenvectors of $X^T X$.

Sub-Section Number :	6
Sub-Section Id :	64065382616
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (C)

MLT_149 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 176 Question Id : 640653577929 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider the following dataset in $\mathbb{R}$:

$$\{-3, \quad -1, \quad 0, \quad 1, \quad 4, \quad 299, \quad 300, \quad 301\}$$

Lloyd's algorithm ($K = 2$) is run on this dataset with the points -10 and 310 as the initial cluster centers. Let a, b be the final cluster centers on convergence, what is the value of the product ab?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

60

✔ **Official Answer Key (Numerical Value): 60**

MLT_150 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 177 Question Id : 640653577930 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset that has 100 data-points, each of which is either a zero or one. The Bernoulli distribution is used to model this data. If the MLE estimate for the parameter p of the Bernoulli distribution is 0.3, how many zeros does the dataset have?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

70

✔ **Official Answer Key (Numerical Value): 70**

MLT_151 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 178 Question Id : 640653577932 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset with 60 points in which all points are either 0 or 1. We use a Bernoulli distribution with the parameter p to model this problem. The prior and posterior distributions for the parameter p are Beta(10,5) and Beta(30,45) respectively. How many data points have the value 0 in this dataset? Note that $p = P(x = 1)$ as usual.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

40

Sub-Section Number :	7
Sub-Section Id :	64065382617
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 40**

MLT_152 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 179 Question Id : 640653577931 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4.5

Question Label : Short Answer Question

Consider a GMM with two components:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7$$

$$\mu_1 = 0, \quad \sigma_1^2 = 1$$

$$\mu_2 = 1, \quad \sigma_2^2 = 2$$

What is the probability that the point $x = 1.2$ comes from the first component? Use the following table for your reference. Here, $\mathcal{N}(x, \mu, \sigma^2)$ is the density of the Gaussian with mean μ and variance σ^2 evaluated at x.

μ	σ^2	$\mathcal{N}(1.2, \mu, \sigma^2)$
0	1	0.194
0	2	0.197
1	1	0.391
1	2	0.279

Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.15 to 0.35

Sub-Section Number :	8
Sub-Section Id :	64065382618
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653577933 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (180 to 181)

✔ Official Answer Key (Numerical Value): 0.15 to 0.35

MLT_153 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 180 Question Id : 640653577934 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the value of a? Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.25 to 1.5

✔ **Official Answer Key (Numerical Value): 1.25 to 1.5**

MLT_154 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 181 Question Id : 640653577935 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the value of b? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.25 to 0.55

✔ **Official Answer Key (Numerical Value): 0.25 to 0.55**

MLT_155 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 6406531297964 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

In a K-means algorithm with $K = 3$, the objective value $F(z_1, \ldots, z_n) = \sum_{i=1}^{n} ||x_i - \mu_{z_i}||^2$ is recorded after each iteration as follows:

Iteration 1: 800, Iteration 2: 500, Iteration 3: 400, Iteration 4: 400, Iteration 5: 400.
At which iteration is the algorithm most likely to reach convergence?

Options :

6406534364889. ✖ 2

6406534364890. ✔ 3

6406534364891. ✖ 5

6406534364892. ✖ 4

Sub-Section Number :	3
Sub-Section Id :	640653206562
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_156 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 131 Question Id : 6406531297965 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a valid kernel function such that $k(x, y) = \phi(x)^T \phi(y)$, where $\phi : \mathbb{R}^d \to \mathbb{R}^D$, where $d < D$. Define

$$k_1(x, y) = \frac{100k(x, y)}{\sqrt{k(x, x)k(y, y)}}.$$

Find $\phi_1 : \mathbb{R}^d \to \mathbb{R}^D$ such that $k_1(x, y) = \phi_1(x)^T \phi_1(y)$.

Note: $||x||_2 = \sqrt{x^T x}$.

Options :

6406534364893. ✖ $\phi_1(x) = \frac{100\phi(x)}{||\phi(x)||_2}$

6406534364894. ✖ $\phi_1(x) = \frac{\phi(x)}{10||\phi(x)||_2}$

6406534364895. ✔ $\phi_1(x) = \frac{10\phi(x)}{||\phi(x)||_2}$

6406534364896. ✖ Such a ϕ_1 does not exist, i.e., k_1 is not a valid kernel.

✔ Official Answer Key: Option (C)

MLT_157 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531297966 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Kernel PCA with a polynomial kernel is applied on the following dataset with three features:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The k-th principal component is given by $w_k = \phi(X)\alpha_k$, where ϕ is the transformation mapping corresponding to the kernel $k(x_i, x_j) = (1 + x_i^T x_j)^2$ for some $\alpha_k \in \mathbb{R}^n$. Find the centered kernel matrix K^c.

Hint: $k_{ij}^c = k_{ij} - \theta_i \mathbb{1}_j^T - \mathbb{1}_i \theta_j^T + P\mathbb{1}_i \mathbb{1}_j^T$, where $\theta_i = \frac{1}{n}\sum_{k=1}^{n} k_{ik}, \forall\, i, P = \frac{1}{n^2}\sum_{i,j} k_{ij}$.

Options :

6406534364897. ✖ $\begin{pmatrix} -1.5 & 1.5 \\ 1.5 & -1.5 \end{pmatrix}$

6406534364898. ✔ $\begin{pmatrix} 1.5 & -1.5 \\ -1.5 & 1.5 \end{pmatrix}$

6406534364899. ✖ $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

6406534364900. ✖ $\begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix}$

Sub-Section Number :	4
Sub-Section Id :	640653206563
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_160 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531297970 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset X of shape (d, n), with $d > n$. The k-th principal component of the dataset can be written as $w_k = \sum_{i=1}^{n} \alpha_i x_i$, where the vector x_i is the i-th data point. The k-th largest eigenvalue and the corresponding eigenvector of $X^T X$ are 10 and $\left[\frac{2}{\sqrt{54}} \quad \frac{3}{\sqrt{54}} \quad \frac{4}{\sqrt{54}} \quad \frac{6}{\sqrt{54}}\right]^T$, respectively. What will be the value of α_2? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.12 to 0.13

✔ **Official Answer Key (Numerical Value): 0.12 to 0.13**

MLT_161 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531297972 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample drawn from a Pareto distribution with the following probability density function:

$$f(x \mid \theta) = \theta x^{-\theta-1}, \quad x \geq 1, \ \theta > 1.$$

You are given the following sample of 5 observations:

$$X = \{2.5,\ 1.8,\ 3.2,\ 2.0,\ 2.7\}.$$

What is the maximum likelihood estimator of θ for the above sample? Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.147 to 1.150

Sub-Section Number :	7
Sub-Section Id :	640653206566
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 1.147 to 1.150**

MLT_162 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531297971 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Consider a dataset $\{x_1, x_2, \ldots, x_{50}\}$, where $x_i \in \mathbb{R}^{1000}$. After performing PCA on the dataset, suppose that the residues become zero after the fifth round. Then, how many numbers do we need to store for this dataset?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5250

Sub-Section Number :	8
Sub-Section Id :	640653206567
Question Shuffling Allowed :	No

Question Id : 6406531297973 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (139 to 141)

✔ **Official Answer Key (Numerical Value): 5250**

MLT_163 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 142 Question Id : 6406531297978 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Fit the linear regression model $y = w\mathbf{x}, w = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^T$ using the squared error.

Options :

6406534364926. ✖ $y = x_1 + 2x_2$

6406534364927. ✔ $y = x_1 - 2x_2$

6406534364928. ✖ $y = x_1 - x_2$

6406534364929. ✖ $y = x_1 + x_2$

✔ **Official Answer Key: Option (B)**

MLT_164 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 6406531298234 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

In a K-means algorithm with $K = 3$, the objective value $F(z_1, \ldots, z_n) = \sum_{i=1}^{n} ||x_i - \mu_{z_i}||^2$ is recorded after each iteration as follows:

Iteration 1: 800, Iteration 2: 500, Iteration 3: 400, Iteration 4: 400, Iteration 5: 400.
At which iteration is the algorithm most likely to reach convergence?

Options :

6406534365660. ✖ 2

6406534365661. ✔ 3

6406534365662. ✖ 5

6406534365663. ✖ 4

Sub-Section Number :	3
Sub-Section Id :	640653206663
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_165 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 131 Question Id : 6406531298235 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a valid kernel function such that $k(x, y) = \phi(x)^T \phi(y)$, where $\phi : \mathbb{R}^d \to \mathbb{R}^D$, where $d < D$. Define

$$k_1(x, y) = \frac{100k(x, y)}{\sqrt{k(x, x)k(y, y)}}.$$

Find $\phi_1 : \mathbb{R}^d \to \mathbb{R}^D$ such that $k_1(x, y) = \phi_1(x)^T \phi_1(y)$.

Note: $||x||_2 = \sqrt{x^T x}$.

Options :

6406534365664. ✖ $\phi_1(x) = \frac{100\phi(x)}{||\phi(x)||_2}$

6406534365665. ✖ $\phi_1(x) = \frac{\phi(x)}{10||\phi(x)||_2}$

6406534365666. ✔ $\phi_1(x) = \frac{10\phi(x)}{||\phi(x)||_2}$

6406534365667. ✖ Such a ϕ_1 does not exist, i.e., k_1 is not a valid kernel.

✔ Official Answer Key: Option (C)

MLT_166 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 132 Question Id : 6406531298236 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Kernel PCA with a polynomial kernel is applied on the following dataset with three features:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The k-th principal component is given by $w_k = \phi(X)\alpha_k$, where ϕ is the transformation mapping corresponding to the kernel $k(x_i, x_j) = (1 + x_i^T x_j)^2$ for some $\alpha_k \in \mathbb{R}^n$. Find the centered kernel matrix K^c.

Hint: $k_{ij}^c = k_{ij} - \theta_i \mathbb{1}_j^T - \mathbb{1}_i \theta_j^T + P\mathbb{1}_i \mathbb{1}_j^T$, where $\theta_i = \frac{1}{n}\sum_{k=1}^{n} k_{ik}, \forall\, i, P = \frac{1}{n^2}\sum_{i,j} k_{ij}$.

Options :

6406534365668. ✖ $\begin{pmatrix} -1.5 & 1.5 \\ 1.5 & -1.5 \end{pmatrix}$

6406534365669. ✔ $\begin{pmatrix} 1.5 & -1.5 \\ -1.5 & 1.5 \end{pmatrix}$

6406534365670. ✖ $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

6406534365671. ✖ $\begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix}$

Sub-Section Number :	4
Sub-Section Id :	640653206664
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLT_169 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 136 Question Id : 6406531298240 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset X of shape (d, n), with $d > n$. The k-th principal component of the dataset can be written as $w_k = \sum_{i=1}^{n} \alpha_i x_i$, where the vector x_i is the i-th data point. The k-th largest eigenvalue and the corresponding eigenvector of $X^T X$ are 10 and $\left[\frac{2}{\sqrt{54}} \quad \frac{3}{\sqrt{54}} \quad \frac{4}{\sqrt{54}} \quad \frac{6}{\sqrt{54}}\right]^T$, respectively. What will be the value of α_2? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.12 to 0.13

✔ **Official Answer Key (Numerical Value): 0.12 to 0.13**

MLT_170 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 137 Question Id : 6406531298242 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample drawn from a Pareto distribution with the following probability density function:

$$f(x \mid \theta) = \theta x^{-\theta-1}, \quad x \geq 1,\ \theta > 1.$$

You are given the following sample of 5 observations:

$$X = \{2.5,\ 1.8,\ 3.2,\ 2.0,\ 2.7\}.$$

What is the maximum likelihood estimator of θ for the above sample? Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.147 to 1.150

Sub-Section Number :	7
Sub-Section Id :	640653206667
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 1.147 to 1.150**

MLT_171 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 138 Question Id : 6406531298241 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Consider a dataset $\{x_1, x_2, \ldots, x_{50}\}$, where $x_i \in \mathbb{R}^{1000}$. After performing PCA on the dataset, suppose that the residues become zero after the fifth round. Then, how many numbers do we need to store for this dataset?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5250

Sub-Section Number :	8
Sub-Section Id :	640653206668
Question Shuffling Allowed :	No

Question Id : 6406531298243 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (139 to 141)

✔ **Official Answer Key (Numerical Value): 5250**

MLT_172 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 142 Question Id : 6406531298248 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Fit the linear regression model $y = wx, w = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^T$ using the squared error.

Options :

6406534365697. ✖ $y = x_1 + 2x_2$

6406534365698. ✔ $y = x_1 - 2x_2$

6406534365699. ✖ $y = x_1 - x_2$

6406534365700. ✖ $y = x_1 + x_2$

✔ Official Answer Key: Option (B)

MLT_173 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 117 Question Id : 640653825110 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the top two principal components of the covariance matrix of a centered dataset. Let $\mathbf{x} \in \mathbb{R}^d$ be a data-point. Which of the following is the reconstruction error of the data-point after projecting it onto the top two principal components?

Options :

6406532775010. ✔ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1 - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532775011. ✖ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1\right|\right|^2 + \left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532775012. ✖ $(\mathbf{x}^T\mathbf{w}_1)^2 + (\mathbf{x}^T\mathbf{w}_2)^2$

6406532775013. ✖ $||\mathbf{x} - \mathbf{w}_1||^2 + ||\mathbf{x} - \mathbf{w}_2||^2$

Sub-Section Number :	3
Sub-Section Id :	640653122750
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLT_175 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 640653825112 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a covariance matrix $\mathbf{C}$ for a mean-centered dataset in $\mathbb{R}^3$. After performing standard PCA, the three principal components turn out to be $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$.

Which of the following statements are true? You can assume that $\mathbf{C}$ is not the zero matrix.

Options :

6406532775018. ✔ $\mathbf{C}$ is a diagonal matrix.

6406532775019. ✔ The diagonal entries of $\mathbf{C}$ are non-negative.

6406532775020. ✖ The diagonal entries of $\mathbf{C}$ are strictly greater than zero.

6406532775021. ✖ $\mathbf{C}$ has to be the identity matrix.

6406532775022. ✖ $\mathbf{C}$ is a matrix of the form $k\mathbf{I}$, where $k > 0$ and $\mathbf{I}$ is the identity matrix.

Sub-Section Number :	5
Sub-Section Id :	640653122752
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_177 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 640653825116 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The EM algorithm is run on a dataset of 100 points modeled using a GMM with three components. Let λ_k^i be the values obtained in the E-step in the final iteration. Here, i is the index of the data-point and k is the index of a component. Also, let $\pi_1 = 0.3, \pi_2 = 0.5, \pi_3 = 0.2$ be the estimates of the mixture probabilities. x_p represent one of the data-points, for some constant p. If $\lambda_1^p + \lambda_3^p = 0.2$, what is the probability that the second component generated x_p given that we have seen the data-point?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.8

Sub-Section Number :	6
Sub-Section Id :	640653122753
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 0.8**

MLT_178 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 122 Question Id : 640653825114 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is a polynomial kernel of degree 4. Find the value of $k\left(\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}\right)$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

✔ **Official Answer Key (Numerical Value): 16**

MLT_179 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 123 Question Id : 640653825115 Question Type : SA
Correct Marks : 4

Question Label : Short Answer Question

Consider the k-means++ algorithm with $k = 2$ for a dataset that has 7 points: $D = \{x_1, \cdots, x_7\}$. The data-point x_7 has been chosen as the first mean, that is, $\mu_1 = x_7$. The Euclidean distance of the remaining points from this mean is given below:

Point	$d(x_i, \mu_1)$
x_1	1
x_2	2
x_3	$\sqrt{2}$
x_4	$\sqrt{3}$
x_5	$\sqrt{5}$
x_6	1

What is the probability of choosing x_2 as the second mean given that x_7 has been chosen as the first mean?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.25

Sub-Section Number : 7
Sub-Section Id : 640653122754
Question Shuffling Allowed : No

Question Id : 640653825117 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (124 to 126)

Question Label : Comprehension

Let $\mathbf{C}$ be the covariance matrix of a mean-centered dataset:

$$\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Standard PCA is performed on this dataset. The first two PCs are given below:

$$\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key (Numerical Value): 0.25

MLT_180 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*
Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 124 Question Id : 640653825118 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Find the variance of the dataset along the first principal component.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_181 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*
Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 125 Question Id : 640653825119 Question Type : SA
Correct Marks : 1
Question Label : Short Answer Question
Find the variance of the dataset along the x-axis.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
2

✔ **Official Answer Key (Numerical Value): 2**

MLT_182 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 640653825120 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Find the coordinates of the point $\begin{bmatrix}2\\2\end{bmatrix}$ in the new coordinate system formed by the principal components.

Options :

6406532775029. ✔ $\begin{bmatrix}2\sqrt{2}\\0\end{bmatrix}$

6406532775030. ✖ $\begin{bmatrix}1\\0\end{bmatrix}$

6406532775031. ✖ $\begin{bmatrix}0\\1\end{bmatrix}$

6406532775032. ✖ $\begin{bmatrix}2\\0\end{bmatrix}$

Sub-Section Number :	8
Sub-Section Id :	640653122755
Question Shuffling Allowed :	No

Question Id : 640653825121 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ **Official Answer Key: Option (A)**

MLT_186 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 640653825129 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If we use the expected value of the posterior as a point-estimate for the parameter of the Bernoulli distribution, what is $\hat{p}$?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.6

Sub-Section Number :	10
Sub-Section Id :	640653122757
Question Shuffling Allowed :	No

Question Id : 640653825124 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (131 to 132)

✔ **Official Answer Key (Numerical Value): 0.6**

MLT_189 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 117 Question Id : 640653827579 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the top two principal components of the covariance matrix of a centered dataset. Let $\mathbf{x} \in \mathbb{R}^d$ be a data-point. Which of the following is the reconstruction error of the data-point after projecting it onto the top two principal components?

Options :

6406532781534. ✔ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1 - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532781535. ✖ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1\right|\right|^2 + \left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532781536. ✖ $(\mathbf{x}^T\mathbf{w}_1)^2 + (\mathbf{x}^T\mathbf{w}_2)^2$

6406532781537. ✖ $||\mathbf{x} - \mathbf{w}_1||^2 + ||\mathbf{x} - \mathbf{w}_2||^2$

Sub-Section Number :	3
Sub-Section Id :	640653123501
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLT_191 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 640653827581 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a covariance matrix $\mathbf{C}$ for a mean-centered dataset in $\mathbb{R}^3$. After performing standard PCA, the three principal components turn out to be $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$.

Which of the following statements are true? You can assume that $\mathbf{C}$ is not the zero matrix.

Options :

6406532781542. ✔ $\mathbf{C}$ is a diagonal matrix.

6406532781543. ✔ The diagonal entries of $\mathbf{C}$ are non-negative.

6406532781544. ✖ The diagonal entries of $\mathbf{C}$ are strictly greater than zero.

6406532781545. ✖ $\mathbf{C}$ has to be the identity matrix.

6406532781546. ✖ $\mathbf{C}$ is a matrix of the form $k\mathbf{I}$, where $k > 0$ and $\mathbf{I}$ is the identity matrix.

Sub-Section Number :	5
Sub-Section Id :	640653123503
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_192 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 640653827585 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The EM algorithm is run on a dataset of 100 points modeled using a GMM with three components. Let λ_k^i be the values obtained in the E-step in the final iteration. Here, i is the index of the data-point and k is the index of a component. Also, let $\pi_1 = 0.3, \pi_2 = 0.5, \pi_3 = 0.2$ be the estimates of the mixture probabilities. x_p represent one of the data-points, for some constant p. If $\lambda_1^p + \lambda_3^p = 0.2$, what is the probability that the second component generated x_p given that we have seen the data-point?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.8

Sub-Section Number :	6
Sub-Section Id :	640653123504
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 0.8**

MLT_193 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 122 Question Id : 640653827583 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is a polynomial kernel of degree 4. Find the value of $k\left(\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}\right)$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

✔ **Official Answer Key (Numerical Value): 16**

MLT_194 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 123 Question Id : 640653827584 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Consider the k-means++ algorithm with $k = 2$ for a dataset that has 7 points: $D = \{x_1, \cdots, x_7\}$. The data-point x_7 has been chosen as the first mean, that is, $\mu_1 = x_7$. The Euclidean distance of the remaining points from this mean is given below:

Point	$d(x_i, \mu_1)$
x_1	1
x_2	2
x_3	$\sqrt{2}$
x_4	$\sqrt{3}$
x_5	$\sqrt{5}$
x_6	1

What is the probability of choosing x_2 as the second mean given that x_7 has been chosen as the first mean?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.25

Sub-Section Number : 7
Sub-Section Id : 640653123505
Question Shuffling Allowed : No

Question Id : 640653827586 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (124 to 126)
Question Label : Comprehension

Let $\mathbf{C}$ be the covariance matrix of a mean-centered dataset:

$$\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Standard PCA is performed on this dataset. The first two PCs are given below:

$$\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (Numerical Value): 0.25

MLT_195 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 640653827589 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Find the coordinates of the point $\begin{bmatrix}2\\2\end{bmatrix}$ in the new coordinate system formed by the principal components.

Options :

6406532781553. ✔ $\begin{bmatrix}2\sqrt{2}\\0\end{bmatrix}$

6406532781554. ✖ $\begin{bmatrix}1\\0\end{bmatrix}$

6406532781555. ✖ $\begin{bmatrix}0\\1\end{bmatrix}$

6406532781556. ✖ $\begin{bmatrix}2\\0\end{bmatrix}$

Sub-Section Number :	8
Sub-Section Id :	640653123506
Question Shuffling Allowed :	No

Question Id : 640653827590 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ **Official Answer Key: Option (A)**

MLT_198 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 640653827598 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If we use the expected value of the posterior as a point-estimate for the parameter of the Bernoulli distribution, what is $\hat{p}$?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.6

Sub-Section Number :	10
Sub-Section Id :	640653123508
Question Shuffling Allowed :	No

Question Id : 640653827593 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (131 to 132)

✔ **Official Answer Key (Numerical Value): 0.6**

MLT_199 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 117 Question Id : 640653827867 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the top two principal components of the covariance matrix of a centered dataset. Let $\mathbf{x} \in \mathbb{R}^d$ be a data-point. Which of the following is the reconstruction error of the data-point after projecting it onto the top two principal components?

Options :

6406532782330. ✔ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1 - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532782331. ✖ $\left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_1)\mathbf{w}_1\right|\right|^2 + \left|\left|\mathbf{x} - (\mathbf{x}^T\mathbf{w}_2)\mathbf{w}_2\right|\right|^2$

6406532782332. ✖ $(\mathbf{x}^T\mathbf{w}_1)^2 + (\mathbf{x}^T\mathbf{w}_2)^2$

6406532782333. ✖ $||\mathbf{x} - \mathbf{w}_1||^2 + ||\mathbf{x} - \mathbf{w}_2||^2$

Sub-Section Number :	3
Sub-Section Id :	640653123608
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLT_201 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 640653827869 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a covariance matrix $\mathbf{C}$ for a mean-centered dataset in $\mathbb{R}^3$. After performing standard PCA, the three principal components turn out to be $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$.

Which of the following statements are true? You can assume that $\mathbf{C}$ is not the zero matrix.

Options :

6406532782338. ✔ $\mathbf{C}$ is a diagonal matrix.

6406532782339. ✔ The diagonal entries of $\mathbf{C}$ are non-negative.

6406532782340. ✖ The diagonal entries of $\mathbf{C}$ are strictly greater than zero.

6406532782341. ✖ $\mathbf{C}$ has to be the identity matrix.

6406532782342. ✖ $\mathbf{C}$ is a matrix of the form $k\mathbf{I}$, where $k > 0$ and $\mathbf{I}$ is the identity matrix.

Sub-Section Number :	5
Sub-Section Id :	640653123610
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_202 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 640653827873 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The EM algorithm is run on a dataset of 100 points modeled using a GMM with three components. Let λ_k^i be the values obtained in the E-step in the final iteration. Here, i is the index of the data-point and k is the index of a component. Also, let $\pi_1 = 0.3, \pi_2 = 0.5, \pi_3 = 0.2$ be the estimates of the mixture probabilities. x_p represent one of the data-points, for some constant p. If $\lambda_1^p + \lambda_3^p = 0.2$, what is the probability that the second component generated x_p given that we have seen the data-point?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.8

Sub-Section Number :	6
Sub-Section Id :	640653123611
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 0.8**

MLT_203 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 122 Question Id : 640653827871 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is a polynomial kernel of degree 4. Find the value of $k\left(\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}\right)$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

✔ **Official Answer Key (Numerical Value): 16**

MLT_204 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 123 Question Id : 640653827872 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Consider the k-means++ algorithm with $k = 2$ for a dataset that has 7 points: $D = \{x_1, \cdots, x_7\}$. The data-point x_7 has been chosen as the first mean, that is, $\mu_1 = x_7$. The Euclidean distance of the remaining points from this mean is given below:

Point	$d(x_i, \mu_1)$
x_1	1
x_2	2
x_3	$\sqrt{2}$
x_4	$\sqrt{3}$
x_5	$\sqrt{5}$
x_6	1

What is the probability of choosing x_2 as the second mean given that x_7 has been chosen as the first mean?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.25

Sub-Section Number : 7
Sub-Section Id : 640653123612
Question Shuffling Allowed : No

Question Id : 640653827874 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (124 to 126)
Question Label : Comprehension

Let $\mathbf{C}$ be the covariance matrix of a mean-centered dataset:

$$\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Standard PCA is performed on this dataset. The first two PCs are given below:

$$\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (Numerical Value): 0.25

MLT_205 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 640653827877 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Find the coordinates of the point $\begin{bmatrix}2\\2\end{bmatrix}$ in the new coordinate system formed by the principal components.

Options :

6406532782349. ✔ $\begin{bmatrix}2\sqrt{2}\\0\end{bmatrix}$

6406532782350. ✖ $\begin{bmatrix}1\\0\end{bmatrix}$

6406532782351. ✖ $\begin{bmatrix}0\\1\end{bmatrix}$

6406532782352. ✖ $\begin{bmatrix}2\\0\end{bmatrix}$

Sub-Section Number :	8
Sub-Section Id :	640653123613
Question Shuffling Allowed :	No

Question Id : 640653827878 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ **Official Answer Key: Option (A)**

MLT_208 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 130 Question Id : 640653827886 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If we use the expected value of the posterior as a point-estimate for the parameter of the Bernoulli distribution, what is $\hat{p}$?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.6

Sub-Section Number :	10
Sub-Section Id :	640653123615
Question Shuffling Allowed :	No

Question Id : 640653827881 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (131 to 132)

✔ **Official Answer Key (Numerical Value): 0.6**

MLT_213 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 151 Question Id : 6406531527884 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535115796. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535115797. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535115798. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535115799. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244940
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_214 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 152 Question Id : 6406531527887 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_215 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 153 Question Id : 6406531527888 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244941
Question Shuffling Allowed :	No

Question Id : 6406531527889 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (154 to 156)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_217 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+20 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 155 Question Id : 6406531527891 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If standard PCA is performed on this dataset, what is the variance along the first principal component? Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4.5

✔ **Official Answer Key (Numerical Value): 4.5**

MLT_218 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+20 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 156 Question Id : 6406531527892 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If standard PCA is performed on this dataset, what is the variance along the second principal component? Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Id : 6406531527896 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (157 to 158)

Question Label : Comprehension

Consider clustering 1D data with a mixture of 2 Gaussians using the EM algorithm. You are given the data points $x_1 = 1, x_2 = 10, x_3 = 20$.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 0.5**

MLT_219 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531527897 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535115821. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115822. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115823. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115824. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_220 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531527898 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244942
Question Shuffling Allowed : No

Question Id : 6406531527893 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (159 to 160)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_227 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 151 Question Id : 6406531528187 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535116639. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535116640. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535116641. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535116642. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653245036
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_228 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 152 Question Id : 6406531528190 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_229 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 153 Question Id : 6406531528191 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653245037
Question Shuffling Allowed :	No

Question Id : 6406531528192 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (154 to 156)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_230 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531528200 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535116664. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535116665. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535116666. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535116667. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_231 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531528201 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653245038
Question Shuffling Allowed : No

Question Id : 6406531528196 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (159 to 160)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_236 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 151 Question Id : 6406531528849 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535118570. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535118571. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535118572. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535118573. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653245221
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_237 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 152 Question Id : 6406531528852 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_238 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 153 Question Id : 6406531528853 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653245222
Question Shuffling Allowed :	No

Question Id : 6406531528854 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (154 to 156)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_239 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531528862 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535118595. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535118596. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535118597. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535118598. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_240 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531528863 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653245223
Question Shuffling Allowed : No

Question Id : 6406531528858 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (159 to 160)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_245 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 160 Question Id : 6406531525177 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535108210. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535108211. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535108212. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535108213. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244053
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_246 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 161 Question Id : 6406531525180 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_247 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 162 Question Id : 6406531525181 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244054
Question Shuffling Allowed :	No

Question Id : 6406531525182 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (163 to 165)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_248 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531525190 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535108235. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535108236. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535108237. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535108238. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_249 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 167 Question Id : 6406531525191 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244055
Question Shuffling Allowed : No

Question Id : 6406531525186 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (168 to 169)

✔ Official Answer Key (Numerical Value): 0.64 to 0.68

MLT_254 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 160 Question Id : 6406531525804 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535109987. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535109988. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535109989. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535109990. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244246
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A), Option (B), Option (D)**

MLT_255 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 161 Question Id : 6406531525807 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_256 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 162 Question Id : 6406531525808 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244247
Question Shuffling Allowed :	No

Question Id : 6406531525809 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (163 to 165)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_257 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531525817 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535110012. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535110013. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535110014. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535110015. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_258 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 167 Question Id : 6406531525818 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244248
Question Shuffling Allowed : No

Question Id : 6406531525813 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (168 to 169)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_263 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 160 Question Id : 6406531526733 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535112632. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535112633. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535112634. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535112635. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244537
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_264 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 161 Question Id : 6406531526736 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_265 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 162 Question Id : 6406531526737 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244538
Question Shuffling Allowed :	No

Question Id : 6406531526738 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (163 to 165)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_266 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531526746 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM?
Here, θ is the set of parameters
$\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535112657. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535112658. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535112659. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535112660. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_267 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 167 Question Id : 6406531526747 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244539
Question Shuffling Allowed : No

Question Id : 6406531526742 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (168 to 169)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_272 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 6406531535772 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535138025. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535138026. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535138027. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535138028. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653247425
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_273 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 120 Question Id : 6406531535775 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_274 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 6406531535776 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653247426
Question Shuffling Allowed :	No

Question Id : 6406531535777 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (122 to 124)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_275 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 125 Question Id : 6406531535785 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535138050. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535138051. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535138052. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535138053. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_276 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 6406531535786 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653247427
Question Shuffling Allowed : No

Question Id : 6406531535781 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ Official Answer Key (Numerical Value): 0.64 to 0.68

MLT_281 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 6406531536112 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following are valid kernel functions? Select all that apply.
Options :

6406535139125. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535139126. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535139127. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535139128. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653247527
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_282 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 120 Question Id : 6406531536115 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_283 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 6406531536116 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653247528
Question Shuffling Allowed :	No

Question Id : 6406531536117 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (122 to 124)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_284 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 125 Question Id : 6406531536125 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535139150. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139151. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139152. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139153. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_285 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 6406531536126 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653247529
Question Shuffling Allowed : No

Question Id : 6406531536121 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ Official Answer Key (Numerical Value): 0.64 to 0.68

MLT_290 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 119 Question Id : 6406531536402 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535139956. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535139957. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535139958. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535139959. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653247629
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_291 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 120 Question Id : 6406531536405 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_292 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 121 Question Id : 6406531536406 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653247630
Question Shuffling Allowed :	No

Question Id : 6406531536407 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (122 to 124)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_293 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 125 Question Id : 6406531536415 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535139981. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139982. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139983. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535139984. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_294 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 126 Question Id : 6406531536416 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653247631
Question Shuffling Allowed : No

Question Id : 6406531536411 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (127 to 128)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_299 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531533285 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535131078. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535131079. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535131080. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535131081. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246688
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_300 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 165 Question Id : 6406531533288 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_301 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531533289 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246689
Question Shuffling Allowed :	No

Question Id : 6406531533290 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (167 to 169)
Question Label : Comprehension

Consider the following 2D dataset shown in the Figure 1. PCA is to be applied on this dataset.

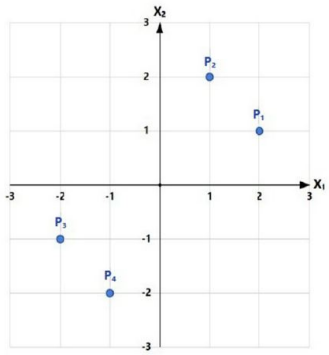


Figure 1: Dataset 1

Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (Numerical Value): 0.45 to 0.50

MLT_302 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 6406531533298 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535131103. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535131104. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535131105. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535131106. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_303 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 6406531533299 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.64 to 0.68

Sub-Section Number : 7

Sub-Section Id : 640653246690

Question Shuffling Allowed : No

Question Id : 6406531533294 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (172 to 173)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_308 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531533610 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535131974. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535131975. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535131976. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535131977. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246784
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_309 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 165 Question Id : 6406531533613 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_310 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531533614 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246785
Question Shuffling Allowed :	No

Question Id : 6406531533615 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (167 to 169)
Question Label : Comprehension

Consider the following 2D dataset shown in the Figure 1. PCA is to be applied on this dataset.

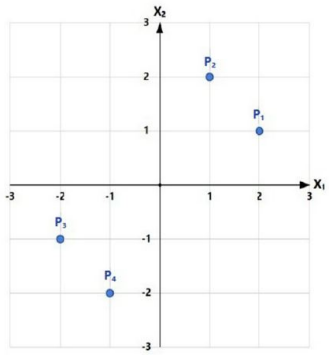


Figure 1: Dataset 1

Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (Numerical Value): 0.45 to 0.50

MLT_311 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 6406531533623 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535131999. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132000. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132001. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132002. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_312 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 6406531533624 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.64 to 0.68

Sub-Section Number : 7

Sub-Section Id : 640653246786

Question Shuffling Allowed : No

Question Id : 6406531533619 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (172 to 173)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_317 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531533935 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535132870. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535132871. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535132872. ✖ $K(x,y) = \|x - y\|, \quad x, y \in \mathbb{R}^n$

6406535132873. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246880
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_318 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 165 Question Id : 6406531533938 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ **Official Answer Key (Numerical Value): 3**

MLT_319 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 166 Question Id : 6406531533939 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246881
Question Shuffling Allowed :	No

Question Id : 6406531533940 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (167 to 169)
Question Label : Comprehension

Consider the following 2D dataset shown in the Figure 1. PCA is to be applied on this dataset.

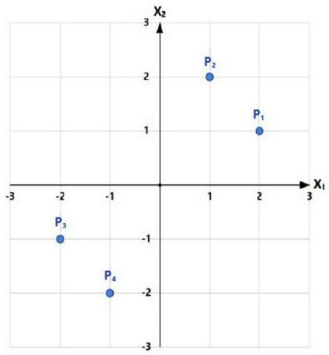


Figure 1: Dataset 1

Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (Numerical Value): 0.45 to 0.50

MLT_320 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 6406531533948 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535132895. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132896. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132897. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535132898. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_321 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 171 Question Id : 6406531533949 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.64 to 0.68

Sub-Section Number :	7
Sub-Section Id :	640653246882
Question Shuffling Allowed :	No

Question Id : 6406531533944 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (172 to 173)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_326 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 31 Question Id : 6406531534118 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535133295. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535133296. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535133297. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535133298. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246926
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_327 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 32 Question Id : 6406531534121 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_328 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 33 Question Id : 6406531534122 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246927
Question Shuffling Allowed :	No

Question Id : 6406531534123 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (34 to 36)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_329 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 37 Question Id : 6406531534131 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535133320. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535133321. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535133322. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535133323. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_330 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 38 Question Id : 6406531534132 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653246928
Question Shuffling Allowed : No

Question Id : 6406531534127 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (39 to 40)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_335 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 31 Question Id : 6406531534845 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535135323. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535135324. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535135325. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535135326. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653247144
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_336 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 32 Question Id : 6406531534848 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_337 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 33 Question Id : 6406531534849 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653247145
Question Shuffling Allowed :	No

Question Id : 6406531534850 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (34 to 36)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_338 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 37 Question Id : 6406531534858 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535135348. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135349. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135350. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135351. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_339 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 38 Question Id : 6406531534859 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653247146
Question Shuffling Allowed : No

Question Id : 6406531534854 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (39 to 40)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_344 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 31 Question Id : 6406531534894 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535135411. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535135412. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535135413. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535135414. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653247158
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_345 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 32 Question Id : 6406531534897 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_346 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 33 Question Id : 6406531534898 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653247159
Question Shuffling Allowed :	No

Question Id : 6406531534899 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (34 to 36)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_347 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 37 Question Id : 6406531534907 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535135436. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135437. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135438. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535135439. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_348 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 38 Question Id : 6406531534908 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653247160
Question Shuffling Allowed : No

Question Id : 6406531534903 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (39 to 40)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_353 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531527057 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535113509. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535113510. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535113511. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535113512. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244641
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_354 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531527060 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_355 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 159 Question Id : 6406531527061 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244642
Question Shuffling Allowed :	No

Question Id : 6406531527062 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (160 to 162)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_356 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 163 Question Id : 6406531527070 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535113534. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535113535. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535113536. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535113537. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_357 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531527071 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244643
Question Shuffling Allowed : No

Question Id : 6406531527066 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (165 to 166)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_362 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531527336 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535114267. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535114268. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535114269. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535114270. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244745
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_363 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531527339 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_364 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 159 Question Id : 6406531527340 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244746
Question Shuffling Allowed :	No

Question Id : 6406531527341 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (160 to 162)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_365 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 163 Question Id : 6406531527349 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535114292. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535114293. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535114294. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535114295. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_366 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531527350 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244747
Question Shuffling Allowed : No

Question Id : 6406531527345 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (165 to 166)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_371 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 157 Question Id : 6406531527615 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535115025. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535115026. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535115027. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535115028. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653244849
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A), Option (B), Option (D)**

MLT_372 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 158 Question Id : 6406531527618 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_373 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 159 Question Id : 6406531527619 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653244850
Question Shuffling Allowed :	No

Question Id : 6406531527620 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (160 to 162)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_374 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 163 Question Id : 6406531527628 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535115050. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115051. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115052. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535115053. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_375 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 164 Question Id : 6406531527629 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653244851
Question Shuffling Allowed : No

Question Id : 6406531527624 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (165 to 166)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_380 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 104 Question Id : 6406531531479 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535125927. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535125928. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535125929. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535125930. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246080
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_381 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 105 Question Id : 6406531531482 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_382 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 106 Question Id : 6406531531483 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246081
Question Shuffling Allowed :	No

Question Id : 6406531531484 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (107 to 109)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_383 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 110 Question Id : 6406531531492 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535125952. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535125953. ✔

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535125954. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535125955. ✖

$$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_384 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 111 Question Id : 6406531531493 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653246082
Question Shuffling Allowed : No

Question Id : 6406531531488 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (112 to 113)

✔ Official Answer Key (Numerical Value): 0.64 to 0.68

MLT_389 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 104 Question Id : 6406531532362 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535128419. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535128420. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535128421. ✖ $K(x,y) = \|x - y\|, \quad x, y \in \mathbb{R}^n$

6406535128422. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246373
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_390 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 105 Question Id : 6406531532365 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_391 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 106 Question Id : 6406531532366 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246374
Question Shuffling Allowed :	No

Question Id : 6406531532367 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (107 to 109)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_392 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 110 Question Id : 6406531532375 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535128444. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535128445. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535128446. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535128447. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_393 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 111 Question Id : 6406531532376 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653246375
Question Shuffling Allowed : No

Question Id : 6406531532371 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (112 to 113)

✔ Official Answer Key (Numerical Value): 0.64 to 0.68

MLT_398 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 104 Question Id : 6406531532925 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are valid kernel functions? Select all that apply.

Options :

6406535129989. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \quad \sigma > 0, x, y \in \mathbb{R}^n$

6406535129990. ✔ $K(x,y) = (x \cdot y + c)^d, \quad c \geq 0, d \in \mathbb{N}, x, y \in \mathbb{R}^n$

6406535129991. ✖ $K(x,y) = \|x-y\|, \quad x, y \in \mathbb{R}^n$

6406535129992. ✔ $K(x,y) = \exp\left(-\frac{\|x-y\|}{\sigma}\right), \quad \sigma > 0, x, y \in \mathbb{R}^n$

Sub-Section Number :	5
Sub-Section Id :	640653246568
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B), Option (D)

MLT_399 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 105 Question Id : 6406531532928 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a mean-centered dataset consisting of 150 patients from the health-care domain with 6 features. The dataset is represented as a 150×6 matrix X. PCA is performed on this dataset, and it is observed that the reconstruction error (residue) vanishes completely after k components. A domain expert provides the following insights about the columns of X, denoted by $c_1, c_2, \ldots, c_6$:

(a) The set $\{c_1, c_2, c_3\}$ is linearly independent.

(b) The following relationships hold:

- $c_4 = 2c_1 - c_2$
- $c_5 = c_2 + c_3$
- $c_6 = 4c_1 - 3c_3$

What is the value of k?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (Numerical Value): 3**

MLT_400 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 106 Question Id : 6406531532929 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset $X \in \mathbb{R}^{d \times n}$ with sample size $n = 100$ (i.e., X has d features and 100 examples), where $d \gg 100$. Suppose the first five eigenvalues of the matrix $X^T X$ are:

$$10,\ 5,\ 3,\ 2,\ 1,$$

and all remaining eigenvalues are zero.

What is the proportion of variance explained along the first principal component? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.45 to 0.50

Sub-Section Number :	6
Sub-Section Id :	640653246569
Question Shuffling Allowed :	No

Question Id : 6406531532930 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (107 to 109)

✔ **Official Answer Key (Numerical Value): 0.45 to 0.50**

MLT_401 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 110 Question Id : 6406531532938 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which among the following represents the likelihood function $L(\theta)$ for GMM? Here, θ is the set of parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \pi_1, \pi_2$.

Options :

6406535130014. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535130015. ✔ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{3} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535130016. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

6406535130017. ✖ $$L(\theta; x_1, x_2, x_3) = \prod_{i=1}^{2} \sum_{k=1}^{2} \pi_k \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(\frac{-1}{2}\left(\frac{x_i - \mu_k}{\sigma_k}\right)^2\right)$$

✔ **Official Answer Key: Option (B)**

MLT_402 | [SA] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 111 Question Id : 6406531532939 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Suppose at the t-th iteration of the EM algorithm, the parameters are:

$\pi_1 = 0.3, \quad \pi_2 = 0.7, \quad \mu_1 = 2, \quad \mu_2 = 3, \quad \sigma_1^2 = 1, \quad \sigma_2^2 = 1.$

The densities of the data points given each mixture component are provided below:

x_i	$f(x_i \mid z_i = 1)$	$f(x_i \mid z_i = 2)$
1	0.242	0.054
10	0.399	0.242
20	0.242	0.399

What is the value of λ_1^1? Enter your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.64 to 0.68

Sub-Section Number : 7
Sub-Section Id : 640653246570
Question Shuffling Allowed : No

Question Id : 6406531532934 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (112 to 113)

✔ **Official Answer Key (Numerical Value): 0.64 to 0.68**

MLT_403 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 168 Question Id : 640653668609 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

In the K-means algorithm, cluster centroids (μ_k) are updated at each iteration to minimize the sum of squared distances between data points and their assigned centroids. The formula used for this update is:

$$\mu_k = \frac{\sum x_i \mathbb{1}(z_i = k)}{\sum \mathbb{1}(z_i = k)}$$

Here:

- x_i represents i^{th} data point in the dataset.
- z_i represents the cluster assignment for data point x_i, where z_i is an integer indicating the cluster index (k).

Suppose you have a dataset of 9 two-dimensional data points and are performing K-means clustering with $K = 3$ clusters. After the first iteration, clusters C_1, C_2, and C_3 have the following data points:

C_1: {(2, 2), (3, 3), (4,4) }

C_2: { (6, 6), (7, 7),(8,8)}

C_3: {(10, 10), (11, 11), (12, 12)}

After the second iteration, which of the following options represents the updated cluster centroid μ_k for C_1 using the provided formula?

Options :

6406532240047. ✖ $\mu 1$ = (2.5, 2.5)

6406532240048. ✖ $\mu 1$ = (4.5, 4.5)

6406532240049. ✔ $\mu 1$ = (3, 3)

6406532240050. ✖ $\mu 1$ = (5.5, 5.5)

✔ Official Answer Key: Option (C)

MLT_405 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653668616 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true regarding applying standard PCA on a centered dataset?

Options :

6406532240065. ✔ The variance of the dataset along the first principal component is maximum.

6406532240066. ✔ Eigenvectors are always orthogonal to each other in PCA.

6406532240067. ✔ The first principal component is the eigenvector corresponding to the largest eigenvalue of the covariance matrix.

6406532240068. ✖ The first principal component is the eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

Sub-Section Number :	4
Sub-Section Id :	64065395192
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_407 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653668615 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset that has 4 points in R^2 that lie on a line passing through the origin:

$$D\left\{\begin{bmatrix}-10\\-30\end{bmatrix},\begin{bmatrix}-5\\-15\end{bmatrix},\begin{bmatrix}5\\15\end{bmatrix},\begin{bmatrix}10\\30\end{bmatrix}\right\}$$

If normal PCA is performed on this dataset, what is the variance along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

625

✔ **Official Answer Key (Numerical Value): 625**

MLT_408 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 173 Question Id : 640653668617 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset of 50 points all of which lie in $\mathbb{R}^5$. The eigenvalues of the covariance matrix are given below:

$$1.7, \quad 1.4, \quad 0.25, \quad 0.20, \quad 0.005$$

If we run the PCA algorithm on this dataset and retain the top-k principal components, what is a good choice of k? (Hint: A good precise mean it captures atleast 95% of variance)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ Official Answer Key (Numerical Value): 4

MLT_409 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 174 Question Id : 640653668619 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the k^{th} mixture after the E-step:

$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 0.5, \lambda_k^4 = 0.5, \lambda_k^5 = 0.3$

What is the estimate of π_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.3 to 0.4

Sub-Section Number :	5
Sub-Section Id :	64065395193
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.3 to 0.4**

MLT_410 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 175 Question Id : 640653668618 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Suppose you have been given a task of estimating the conversion rate of google click of new online advertisement campaign. You have collected data from a limited sample of 50 users, where 15 of them have converted. You want to use Bayesian estimation with a prior distribution to provide a more robust estimate of the conversion rate.

Assume you have prior information and you decide to use a beta distribution as your prior. You choose a beta distribution with parameters $\alpha = 3$ and $\beta = 4$ to capture your prior beliefs.

Calculate the posterior mean?

Enter your answer correctly up to two decimals

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.2 to 0.3

Sub-Section Number :	6
Sub-Section Id :	64065395194
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653668611 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (176 to 177)

✔ Official Answer Key (Numerical Value): 0.2 to 0.3

MLT_412 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 178 Question Id : 640653668621 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true?

Options :

6406532240072. ✖ The eigenvectors of C and K are related by a scaling factor $r \in \mathbb{R}$

6406532240073. ✔ Computing eigenvectors using the matrix K reduces number of computations significantly than using the matrix C

6406532240074. ✔ Suppose the maximum eigenvalue of C is λ_1, then the maximum eigenvalue of XX^T is $n\lambda_1$

6406532240075. ✖ The eigenvectors of $C \in \mathbb{R}^d$ and $K \in \mathbb{R}^d$

6406532240076. ✔ The eigenvalues of both C and K are non-negative

✔ Official Answer Key: Option (B), Option (C), Option (E)

MLT_415 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 168 Question Id : 640653668970 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

In the K-means algorithm, cluster centroids (μ_k) are updated at each iteration to minimize the sum of squared distances between data points and their assigned centroids. The formula used for this update is:

$$\mu_k = \frac{\sum x_i \mathbb{1}(z_i = k)}{\sum \mathbb{1}(z_i = k)}$$

Here:

- x_i represents i^{th} data point in the dataset.
- z_i represents the cluster assignment for data point x_i, where z_i is an integer indicating the cluster index (k).

Suppose you have a dataset of 9 two-dimensional data points and are performing K-means clustering with $K = 3$ clusters. After the first iteration, clusters C_1, C_2, and C_3 have the following data points:

C_1: {(2, 2), (3, 3), (4,4) }

C_2: { (6, 6), (7, 7),(8,8)}

C_3: {(10, 10), (11, 11), (12, 12)}

After the second iteration, which of the following options represents the updated cluster centroid μ_k for C_1 using the provided formula?

Options :

6406532241143. ✖ $\mu 1$ = (2.5, 2.5)

6406532241144. ✖ $\mu 1$ = (4.5, 4.5)

6406532241145. ✔ $\mu 1$ = (3, 3)

6406532241146. ✖ $\mu 1$ = (5.5, 5.5)

✔ Official Answer Key: Option (C)

MLT_417 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653668977 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true regarding applying standard PCA on a centered dataset?

Options :

6406532241161. ✔ The variance of the dataset along the first principal component is maximum.

6406532241162. ✔ Eigenvectors are always orthogonal to each other in PCA.

6406532241163. ✔ The first principal component is the eigenvector corresponding to the largest eigenvalue of the covariance matrix.

6406532241164. ✖ The first principal component is the eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

Sub-Section Number :	4
Sub-Section Id :	64065395297
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_418 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653668976 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset that has 4 points in R^2 that lie on a line passing through the origin:

$$D\left\{\begin{bmatrix}-10\\-30\end{bmatrix},\begin{bmatrix}-5\\-15\end{bmatrix},\begin{bmatrix}5\\15\end{bmatrix},\begin{bmatrix}10\\30\end{bmatrix}\right\}$$

If normal PCA is performed on this dataset, what is the variance along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

625

✔ **Official Answer Key (Numerical Value): 625**

MLT_419 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 173 Question Id : 640653668978 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset of 50 points all of which lie in $\mathbb{R}^5$. The eigenvalues of the covariance matrix are given below:

$$1.7, \quad 1.4, \quad 0.25, \quad 0.20, \quad 0.005$$

If we run the PCA algorithm on this dataset and retain the top-k principal components, what is a good choice of k? (Hint: A good precise mean it captures atleast 95% of variance)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ Official Answer Key (Numerical Value): 4

MLT_420 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 174 Question Id : 640653668980 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the k^{th} mixture after the E-step:

$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 0.5, \lambda_k^4 = 0.5, \lambda_k^5 = 0.3$

What is the estimate of π_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.3 to 0.4

Sub-Section Number :	5
Sub-Section Id :	64065395298
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.3 to 0.4**

MLT_421 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 175 Question Id : 640653668979 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Suppose you have been given a task of estimating the conversion rate of google click of new online advertisement campaign. You have collected data from a limited sample of 50 users, where 15 of them have converted. You want to use Bayesian estimation with a prior distribution to provide a more robust estimate of the conversion rate.

Assume you have prior information and you decide to use a beta distribution as your prior. You choose a beta distribution with parameters $\alpha = 3$ and $\beta = 4$ to capture your prior beliefs.

Calculate the posterior mean?

Enter your answer correctly up to two decimals

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.2 to 0.3

Sub-Section Number :	6
Sub-Section Id :	64065395299
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653668972 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (176 to 177)

✔ Official Answer Key (Numerical Value): 0.2 to 0.3

MLT_422 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 178 Question Id : 640653668982 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true?

Options :

6406532241168. ✖ The eigenvectors of C and K are related by a scaling factor $r \in \mathbb{R}$

6406532241169. ✔ Computing eigenvectors using the matrix K reduces number of computations significantly than using the matrix C

6406532241170. ✔ Suppose the maximum eigenvalue of C is λ_1, then the maximum eigenvalue of XX^T is $n\lambda_1$

6406532241171. ✖ The eigenvectors of $C \in \mathbb{R}^d$ and $K \in \mathbb{R}^d$

6406532241172. ✔ The eigenvalues of both C and K are non-negative

✔ Official Answer Key: Option (B), Option (C), Option (E)

MLT_423 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 168 Question Id : 640653669231 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

In the K-means algorithm, cluster centroids (μ_k) are updated at each iteration to minimize the sum of squared distances between data points and their assigned centroids. The formula used for this update is:

$$\mu_k = \frac{\sum x_i \mathbb{1}(z_i = k)}{\sum \mathbb{1}(z_i = k)}$$

Here:

- x_i represents i^{th} data point in the dataset.
- z_i represents the cluster assignment for data point x_i, where z_i is an integer indicating the cluster index (k).

Suppose you have a dataset of 9 two-dimensional data points and are performing K-means clustering with $K = 3$ clusters. After the first iteration, clusters C_1, C_2, and C_3 have the following data points:

C_1: {(2, 2), (3, 3), (4,4) }

C_2: { (6, 6), (7, 7),(8,8)}

C_3: {(10, 10), (11, 11), (12, 12)}

After the second iteration, which of the following options represents the updated cluster centroid μ_k for C_1 using the provided formula?

Options :

6406532241895. ✖ $\mu 1$ = (2.5, 2.5)

6406532241896. ✖ $\mu 1$ = (4.5, 4.5)

6406532241897. ✔ $\mu 1$ = (3, 3)

6406532241898. ✖ $\mu 1$ = (5.5, 5.5)

✔ Official Answer Key: Option (C)

MLT_425 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 170 Question Id : 640653669238 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true regarding applying standard PCA on a centered dataset?

Options :

6406532241913. ✔ The variance of the dataset along the first principal component is maximum.

6406532241914. ✔ Eigenvectors are always orthogonal to each other in PCA.

6406532241915. ✔ The first principal component is the eigenvector corresponding to the largest eigenvalue of the covariance matrix.

6406532241916. ✖ The first principal component is the eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

Sub-Section Number :	4
Sub-Section Id :	64065395396
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (B), Option (C)

MLT_426 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 172 Question Id : 640653669237 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset that has 4 points in R^2 that lie on a line passing through the origin:

$$D\left\{\begin{bmatrix}-10\\-30\end{bmatrix},\begin{bmatrix}-5\\-15\end{bmatrix},\begin{bmatrix}5\\15\end{bmatrix},\begin{bmatrix}10\\30\end{bmatrix}\right\}$$

If normal PCA is performed on this dataset, what is the variance along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

625

✔ **Official Answer Key (Numerical Value): 625**

MLT_427 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 173 Question Id : 640653669239 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset of 50 points all of which lie in $\mathbb{R}^5$. The eigenvalues of the covariance matrix are given below:

$$1.7, \quad 1.4, \quad 0.25, \quad 0.20, \quad 0.005$$

If we run the PCA algorithm on this dataset and retain the top-k principal components, what is a good choice of k? (Hint: A good precise mean it captures atleast 95% of variance)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ Official Answer Key (Numerical Value): 4

MLT_428 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 174 Question Id : 640653669241 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the k^{th} mixture after the E-step:

$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 0.5, \lambda_k^4 = 0.5, \lambda_k^5 = 0.3$

What is the estimate of π_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.3 to 0.4

Sub-Section Number :	5
Sub-Section Id :	64065395397
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 0.3 to 0.4**

MLT_429 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 175 Question Id : 640653669240 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Suppose you have been given a task of estimating the conversion rate of google click of new online advertisement campaign. You have collected data from a limited sample of 50 users, where 15 of them have converted. You want to use Bayesian estimation with a prior distribution to provide a more robust estimate of the conversion rate.

Assume you have prior information and you decide to use a beta distribution as your prior. You choose a beta distribution with parameters $\alpha = 3$ and $\beta = 4$ to capture your prior beliefs.

Calculate the posterior mean?

Enter your answer correctly up to two decimals

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.2 to 0.3

Sub-Section Number :	6
Sub-Section Id :	64065395398
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653669233 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (176 to 177)

✔ Official Answer Key (Numerical Value): 0.2 to 0.3

MLT_430 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 178 Question Id : 640653669243 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true?

Options :

6406532241920. ✖ The eigenvectors of C and K are related by a scaling factor $r \in \mathbb{R}$

6406532241921. ✔ Computing eigenvectors using the matrix K reduces number of computations significantly than using the matrix C

6406532241922. ✔ Suppose the maximum eigenvalue of C is λ_1, then the maximum eigenvalue of XX^T is $n\lambda_1$

6406532241923. ✖ The eigenvectors of $C \in \mathbb{R}^d$ and $K \in \mathbb{R}^d$

6406532241924. ✔ The eigenvalues of both C and K are non-negative

✔ Official Answer Key: Option (B), Option (C), Option (E)

MLT_431 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 15 Question Id : 640653995442 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. Define the functions k_1 and k_2 as:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + (x_1 + x_2)(y_1 + y_2)$$
$$k_2(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + (x_1^2 + y_2^2) + 3$$

Which of the following statements is true?

Options :

6406533360624. ✖ Both k_1 and k_2 are valid kernels.

6406533360625. ✔ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406533360626. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406533360627. ✖ Neither k_1 nor k_2 is a valid kernel.

✔ Official Answer Key: Option (B)

MLT_432 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 16 Question Id : 640653995444 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \rightarrow R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406533360630. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406533360631. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406533360632. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406533360633. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

Sub-Section Number :	3
Sub-Section Id :	640653146776
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_433 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 17 Question Id : 640653995443 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given n data points in a d-dimensional space with a non-linear relationship, we apply kernel PCA to reduce the dimensionality and select the first k principal components. Is it possible for k to be greater than d?

Options :

6406533360628. ✔ yes

6406533360629. ✖ No

Sub-Section Number :	4
Sub-Section Id :	640653146777
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_434 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 18 Question Id : 640653995440 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following expressions is the reconstruction error for a dataset of n points, with respect to a line passing through the origin represented by the vector $\mathbf{w}$. Note that $||\mathbf{w}|| = 1$.

Options :

6406533360619. ✔ $\frac{1}{n}\sum_{i=1}^{n} ||\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}||^2$

6406533360620. ✔ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]^T \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]$

6406533360621. ✖ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i^T\mathbf{x}_i \mid (\mathbf{x}_i^T\mathbf{w})^2\right]$

6406533360622. ✖ $-\frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i^T\mathbf{w})^2$

Sub-Section Number :	5
Sub-Section Id :	640653146778
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_435 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 19 Question Id : 640653995449 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $X_1, X_2, \ldots, X_n$ be n i.i.d. samples with parameter θ, which follows one of the following PDFs:

For $\theta = -1$, we have

$$f(x \mid \theta) = \begin{cases} 5x^4, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

For $\theta = 1$, we have

$$f(x \mid \theta) = \begin{cases} 1, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Suppose we wish to find the maximum likelihood estimate of θ, then which among the following are true?

Options :

6406533360643. ✔ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = 1$

6406533360644. ✖ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = 1$

6406533360645. ✖ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = -1$

6406533360646. ✔ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = -1$

Sub-Section Number :	6
Sub-Section Id :	640653146779
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_436 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 20 Question Id : 640653995441 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The eigenvalues of the covariance matrix of a centered dataset in $\mathbb{R}^5$ are $30, 10, 10, 0, 0$. Standard PCA is performed on this dataset. What is the variance captured by the top two principal components expressed as a percentage of total variance?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

Sub-Section Number :	7
Sub-Section Id :	640653146780
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 80**

MLT_437 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 21 Question Id : 640653995448 Question Type : SA
Correct Marks : 5
Question Label : Short Answer Question

Consider the following data points for k-means clustering.

$$(-1,0),(-1,1),(-1,-1),(2,0),(3,1),(3,-1),(4,0)$$

In the initialization step of k-means with k = 2, suppose $\mu_1^0 = (-1,0)$ and $\mu_2^0 = (2,0)$. Distances of datapoints from initial cluster means is tabulated below:

x_i	$\|x_i - \mu_1^0\|_2^2$	$\|x_i - \mu_2^0\|_2^2$
$(-1,0)$	0	3
$(-1,1)$	1	10
$(-1,-1)$	1	10
$(2,0)$	3	0
$(3,1)$	17	2
$(3,-1)$	17	2
$(4,0)$	5	2

As per these cluster centers, the data points are then assigned to either cluster 1 or cluster 2. After this assignment, what will be the value of the objective function?
Note: Objective function is given by

$$F(z_1, z_2, \ldots, z_n) = \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|_2^2$$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
6

Sub-Section Number :	8
Sub-Section Id :	640653146781
Question Shuffling Allowed :	Yes

✔ Official Answer Key (Numerical Value): 6

MLT_438 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 22 Question Id : 640653995450 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a GMM for 5 points:

$$x_1 = 1, x_2 = 1.2, x_3 = 2, x_4 = 1.5, x_5 = 0.5$$

At some time-step in the EM algorithm, following are the values of λ_k^i for the k-th mixture after the E-step:

$$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 2.5, \lambda_k^4 = 0.6, \lambda_k^5 = 0.8$$

What is the estimate of μ_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.50 to 1.60

✔ **Official Answer Key (Numerical Value): 1.50 to 1.60**

MLT_439 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 24 Question Id : 640653995438 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the length of the projection of $\mathbf{x}$ onto the line defined by $\mathbf{w}$.
Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.1

✔ **Official Answer Key (Numerical Value): 4.8 to 5.1**

MLT_440 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 25 Question Id : 640653995439 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Calculate the magnitude(norm) of reconstruction error after projecting $\mathbf{x}$ onto the line defined by $\mathbf{w}$.

Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.5 to 0.9

Question Id : 640653995445 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (26 to 27)

Question Label : Comprehension

A k-means++ algorithm with $k = 3$ is applied on the following 2D points:

$$(0,1), (1,0), (1,2), (2,1), (2,3), (2,4), (3,2)$$

First cluster mean μ_1^0 is chosen as (2, 1).

Suppose the point with the highest score is chosen as the 2nd cluster mean μ_2^0.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 0.5 to 0.9**

MLT_441 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 26 Question Id : 640653995446 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What is μ_2^0? Use squared distance to calculate the scores.

Options :

6406533360634. ✖ (0,1)

6406533360635. ✖ (2,3)

6406533360636. ✖ (3,2)

6406533360637. ✔ (2,4)

✔ Official Answer Key: Option (D)

MLT_443 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 15 Question Id : 640653995724 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Let $\mathrm{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathrm{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. Define the functions k_1 and k_2 as:

$$k_1(\mathrm{x}, \mathrm{y}) = x_1y_1 + x_2y_2 + (x_1 + x_2)(y_1 + y_2)$$
$$k_2(\mathrm{x}, \mathrm{y}) = x_1y_1 + x_2y_2 + (x_1^2 + y_2^2) + 3$$

Which of the following statements is true?

Options :

6406533361408. ✖ Both k_1 and k_2 are valid kernels.

6406533361409. ✔ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406533361410. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406533361411. ✖ Neither k_1 nor k_2 is a valid kernel.

✔ Official Answer Key: Option (B)

MLT_444 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 16 Question Id : 640653995726 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \rightarrow R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406533361414. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406533361415. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406533361416. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406533361417. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

Sub-Section Number :	3
Sub-Section Id :	640653146865
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_445 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 17 Question Id : 640653995725 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given n data points in a d-dimensional space with a non-linear relationship, we apply kernel PCA to reduce the dimensionality and select the first k principal components. Is it possible for k to be greater than d?

Options :

6406533361412. ✔ yes

6406533361413. ✖ No

Sub-Section Number :	4
Sub-Section Id :	640653146866
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_446 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 18 Question Id : 640653995722 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following expressions is the reconstruction error for a dataset of n points, with respect to a line passing through the origin represented by the vector $\mathbf{w}$. Note that $||\mathbf{w}|| = 1$.

Options :

6406533361403. ✔ $\frac{1}{n}\sum_{i=1}^{n} ||\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}||^2$

6406533361404. ✔ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]^T \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]$

6406533361405. ✖ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i^T\mathbf{x}_i \mid (\mathbf{x}_i^T\mathbf{w})^2\right]$

6406533361406. ✖ $-\frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i^T\mathbf{w})^2$

Sub-Section Number :	5
Sub-Section Id :	640653146867
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_447 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 19 Question Id : 640653995731 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $X_1, X_2, \ldots, X_n$ be n i.i.d. samples with parameter θ, which follows one of the following PDFs:

For $\theta = -1$, we have

$$f(x \mid \theta) = \begin{cases} 5x^4, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

For $\theta = 1$, we have

$$f(x \mid \theta) = \begin{cases} 1, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Suppose we wish to find the maximum likelihood estimate of θ, then which among the following are true?

Options :

6406533361427. ✔ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = 1$

6406533361428. ✖ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = 1$

6406533361429. ✖ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = -1$

6406533361430. ✔ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = -1$

Sub-Section Number :	6
Sub-Section Id :	640653146868
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_448 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 20 Question Id : 640653995723 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The eigenvalues of the covariance matrix of a centered dataset in $\mathbb{R}^5$ are $30, 10, 10, 0, 0$. Standard PCA is performed on this dataset. What is the variance captured by the top two principal components expressed as a percentage of total variance?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

Sub-Section Number :	7
Sub-Section Id :	640653146869
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 80**

MLT_449 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 21 Question Id : 640653995730 Question Type : SA
Correct Marks : 5
Question Label : Short Answer Question

Consider the following data points for k-means clustering.

$$(-1,0),(-1,1),(-1,-1),(2,0),(3,1),(3,-1),(4,0)$$

In the initialization step of k-means with k = 2, suppose $\mu_1^0 = (-1,0)$ and $\mu_2^0 = (2,0)$. Distances of datapoints from initial cluster means is tabulated below:

x_i	$\|x_i - \mu_1^0\|_2^2$	$\|x_i - \mu_2^0\|_2^2$
$(-1,0)$	0	3
$(-1,1)$	1	10
$(-1,-1)$	1	10
$(2,0)$	3	0
$(3,1)$	17	2
$(3,-1)$	17	2
$(4,0)$	5	2

As per these cluster centers, the data points are then assigned to either cluster 1 or cluster 2. After this assignment, what will be the value of the objective function?
Note: Objective function is given by

$$F(z_1, z_2, \ldots, z_n) = \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|_2^2$$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
6

Sub-Section Number :	8
Sub-Section Id :	640653146870
Question Shuffling Allowed :	Yes

✔ Official Answer Key (Numerical Value): 6

MLT_450 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 22 Question Id : 640653995732 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a GMM for 5 points:

$$x_1 = 1, x_2 = 1.2, x_3 = 2, x_4 = 1.5, x_5 = 0.5$$

At some time-step in the EM algorithm, following are the values of λ_k^i for the k-th mixture after the E-step:

$$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 2.5, \lambda_k^4 = 0.6, \lambda_k^5 = 0.8$$

What is the estimate of μ_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.50 to 1.60

✔ **Official Answer Key (Numerical Value): 1.50 to 1.60**

MLT_451 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 24 Question Id : 640653995720 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the length of the projection of $\mathbf{x}$ onto the line defined by $\mathbf{w}$. Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.1

✔ **Official Answer Key (Numerical Value): 4.8 to 5.1**

MLT_452 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 26 Question Id : 640653995728 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What is μ_2^0? Use squared distance to calculate the scores.

Options :

6406533361418. ✖ (0,1)

6406533361419. ✖ (2,3)

6406533361420. ✖ (3,2)

6406533361421. ✔ (2,4)

✔ **Official Answer Key: Option (D)**

MLT_453 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 15 Question Id : 640653996006 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. Define the functions k_1 and k_2 as:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + (x_1 + x_2)(y_1 + y_2)$$
$$k_2(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + (x_1^2 + y_2^2) + 3$$

Which of the following statements is true?

Options :

6406533362192. ✖ Both k_1 and k_2 are valid kernels.

6406533362193. ✔ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406533362194. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406533362195. ✖ Neither k_1 nor k_2 is a valid kernel.

✔ Official Answer Key: Option (B)

MLT_454 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 16 Question Id : 640653996008 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following kernel:

$$k : R^2 \times R^2 \rightarrow R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406533362198. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$

6406533362199. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$

6406533362200. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$

6406533362201. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

Sub-Section Number :	3
Sub-Section Id :	640653146954
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_455 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 17 Question Id : 640653996007 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given n data points in a d-dimensional space with a non-linear relationship, we apply kernel PCA to reduce the dimensionality and select the first k principal components. Is it possible for k to be greater than d?

Options :

6406533362196. ✔ yes

6406533362197. ✖ No

Sub-Section Number :	4
Sub-Section Id :	640653146955
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_456 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 18 Question Id : 640653996004 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following expressions is the reconstruction error for a dataset of n points, with respect to a line passing through the origin represented by the vector $\mathbf{w}$. Note that $||\mathbf{w}|| = 1$.

Options :

6406533362187. ✔ $\frac{1}{n}\sum_{i=1}^{n} ||\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}||^2$

6406533362188. ✔ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]^T \left[\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\right]$

6406533362189. ✖ $\frac{1}{n}\sum_{i=1}^{n} \left[\mathbf{x}_i^T\mathbf{x}_i \mid (\mathbf{x}_i^T\mathbf{w})^2\right]$

6406533362190. ✖ $-\frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i^T\mathbf{w})^2$

Sub-Section Number :	5
Sub-Section Id :	640653146956
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

MLT_457 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 19 Question Id : 640653996013 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $X_1, X_2, \ldots, X_n$ be n i.i.d. samples with parameter θ, which follows one of the following PDFs:

For $\theta = -1$, we have

$$f(x \mid \theta) = \begin{cases} 5x^4, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

For $\theta = 1$, we have

$$f(x \mid \theta) = \begin{cases} 1, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Suppose we wish to find the maximum likelihood estimate of θ, then which among the following are true?

Options :

6406533362211. ✔ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = 1$

6406533362212. ✖ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = 1$

6406533362213. ✖ If $\prod_{i=1}^{n} 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = -1$

6406533362214. ✔ If $\prod_{i=1}^{n} 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = -1$

Sub-Section Number :	6
Sub-Section Id :	640653146957
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_458 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 20 Question Id : 640653996005 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The eigenvalues of the covariance matrix of a centered dataset in $\mathbb{R}^5$ are $30, 10, 10, 0, 0$. Standard PCA is performed on this dataset. What is the variance captured by the top two principal components expressed as a percentage of total variance?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

Sub-Section Number :	7
Sub-Section Id :	640653146958
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 80**

MLT_459 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 21 Question Id : 640653996012 Question Type : SA
Correct Marks : 5
Question Label : Short Answer Question

Consider the following data points for k-means clustering.

$$(-1,0),(-1,1),(-1,-1),(2,0),(3,1),(3,-1),(4,0)$$

In the initialization step of k-means with k = 2, suppose $\mu_1^0 = (-1,0)$ and $\mu_2^0 = (2,0)$. Distances of datapoints from initial cluster means is tabulated below:

x_i	$\|x_i - \mu_1^0\|_2^2$	$\|x_i - \mu_2^0\|_2^2$
$(-1,0)$	0	3
$(-1,1)$	1	10
$(-1,-1)$	1	10
$(2,0)$	3	0
$(3,1)$	17	2
$(3,-1)$	17	2
$(4,0)$	5	2

As per these cluster centers, the data points are then assigned to either cluster 1 or cluster 2. After this assignment, what will be the value of the objective function?
Note: Objective function is given by

$$F(z_1, z_2, \ldots, z_n) = \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|_2^2$$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
6

Sub-Section Number :	8
Sub-Section Id :	640653146959
Question Shuffling Allowed :	Yes

✔ Official Answer Key (Numerical Value): 6

MLT_460 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 22 Question Id : 640653996014 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a GMM for 5 points:

$$x_1 = 1, x_2 = 1.2, x_3 = 2, x_4 = 1.5, x_5 = 0.5$$

At some time-step in the EM algorithm, following are the values of λ_k^i for the k-th mixture after the E-step:

$$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 2.5, \lambda_k^4 = 0.6, \lambda_k^5 = 0.8$$

What is the estimate of μ_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.50 to 1.60

✔ **Official Answer Key (Numerical Value): 1.50 to 1.60**

MLT_461 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 24 Question Id : 640653996002 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the length of the projection of $\mathbf{x}$ onto the line defined by $\mathbf{w}$. Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.1

✔ **Official Answer Key (Numerical Value): 4.8 to 5.1**

MLT_462 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › PCA Fundamentals & Projection

Question Number : 26 Question Id : 640653996010 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What is μ_2^0? Use squared distance to calculate the scores.

Options :

6406533362202. ✖ (0,1)

6406533362203. ✖ (2,3)

6406533362204. ✖ (3,2)

6406533362205. ✔ (2,4)

✔ **Official Answer Key: Option (D)**

• **Representation Learning & Evaluation Metrics** **(12 Questions)**

MLT_004 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 173 Question Id : 640653345309 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the value of the accuracy. Write your answer correct upto 2 decimal places, do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.55 to 0.57

✔ **Official Answer Key (Numerical Value): 0.55 to 0.57**

MLT_005 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 174 Question Id : 640653345310 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the value of the precision. Write your answer correct upto 2 decimal places, do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.43 to 0.45

✔ **Official Answer Key (Numerical Value): 0.43 to 0.45**

MLT_006 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 175 Question Id : 640653345311 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the value of the recall. Write your answer correct upto 2 decimal places, do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.515 to 0.535

✔ **Official Answer Key (Numerical Value): 0.515 to 0.535**

MLT_007 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 176 Question Id : 640653345312 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Find the value of the F-1 score. Write your answer correct upto 2 decimal places, do not round up or off.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.460 to 0.485

Question Id : 640653345323 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (177 to 178)

Question Label : Comprehension

Ajay has a dataset that has a non-linear relationship between the features and the label. The dimension of training samples is 4×2 (excluding the additional all-ones feature for the bias term, which you should add), label vector is 4 × 1 and weight vector is 3 × 1.

X	y
[1 , 1]	1
[2 , 2]	1
[3 , 2]	2
[4 , 3]	2

w
0.1
0.2
0.3

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key (Numerical Value): 0.460 to 0.485

MLT_008 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 177 Question Id : 640653345324 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Suppose Ajay has to compute the ridge loss value for this data with regularization parameter value as 0.2 for performing ridge regression. What will be the value of ridge loss correct up to 2 decimal places?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.35 to 0.37

✔ **Official Answer Key (Numerical Value): 0.35 to 0.37**

MLT_009 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 178 Question Id : 640653345325 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

For the same dataset, suppose Ajay has to compute the lasso loss value with regularization parameter value as 0.2 for performing lasso regression. What will be the value of lasso loss correct up to 2 decimal places?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.39 to 0.43

Question Id : 640653345331 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (179 to 180)

✔ **Official Answer Key (Numerical Value): 0.39 to 0.43**

MLT_010 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 179 Question Id : 640653345332 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What will be the prediction for an example having the features vector as (2, 1)? Assume the classes are 1 and −1.

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-1

✔ **Official Answer Key (Numerical Value): -1**

MLT_011 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 180 Question Id : 640653345333 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Find the accuracy for the given test data-set. Write your answer correct to two decimal places.

x_1	x_2	True label (y)
1	−2	−1
−2	−2	−1
1	2	1
3	2	1

Test data-set

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.73 to 0.77

Sub-Section Number :	4
Sub-Section Id :	64065348797
Question Shuffling Allowed :	No

Question Id : 640653345313 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (181 to 182)

✔ Official Answer Key (Numerical Value): 0.73 to 0.77

MLT_013 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 182 Question Id : 640653345315 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the output of following code snippet?

```
print(C.sum().sum())
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

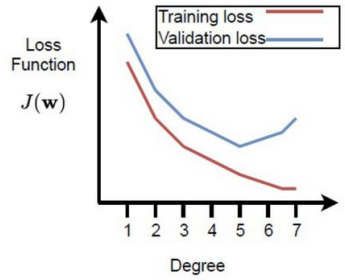
58

Question Id : 640653345326 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (183 to 184)

Question Label : Comprehension

Abhishek trains a polynomial regression model on a dataset. He tries with different degree of polynomial transformation and gets following learning curve:



Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key (Numerical Value): 58

MLT_014 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 183 Question Id : 640653345327 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What value of degree of polynomial transformation should Abhishek choose for the given dataset?

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ **Official Answer Key (Numerical Value): 5**

MLT_021 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 190 Question Id : 640653345329 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

In ridge regression, the loss function is computed as $J(\mathbf{w}) = L + \lambda P$ Where P is penalty term for ridge regularization, λ is rate of regularization and L is plain loss function without any regularization. Choose the correct statements from the following:

Options :

6406531148594. ✔ As λ tends to infinite, the solution of ridge regularization tends towards origin.

6406531148595. ✖ As λ tends to zero, the solution of ridge regularization tends towards origin.

6406531148596. ✖ As λ tends to infinite, the solution of ridge regularization tends towards the solution of least squares.

6406531148597. ✔ As λ tends to zero, the solution of ridge regularization tends towards the solution of least squares.

6406531148598. ✔ The penalty term P represents circles if we draw the contour plot of ridge loss function.

6406531148599. ✖ The penalty term P represents squares if we draw the contour plot of ridge loss function.

Sub-Section Number :	7
Sub-Section Id :	64065348800
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D), Option (E)

MLT_023 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Representation Learning & Evaluation Metrics

Question Number : 192 Question Id : 640653345322 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Ankita had a feature vector **x** and she wanted to apply polynomial transformation on **x**. She had written a code snippet as follows:

```
import itertools, functools
import numpy as np
x = np.array([[2, 3]])
degree= 3
x_t = x.transpose()
features = [np.ones(len(x))]
for degree in range(1, degree + 1):
  for items in itertools.combinations_with_replacement(x_t, degree):
    features.append(functools.reduce(lambda x, y: x * y, items))
output = np.sum(np.asarray(features).transpose())
print(int(output))
```

What is the output of the code snippet?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

90

✔ **Official Answer Key (Numerical Value): 90**

• Covariance Matrix & Eigen-decomposition (7 Questions)

MLT_038 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 139 Question Id : 6406531298514 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following represents the covariance matrix C?

Options :

6406534366459. ✖ $\frac{1}{3}\begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix}$

6406534366460. ✔ $\frac{1}{3}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$

6406534366461. ✖ $\frac{1}{3}\begin{bmatrix}0 & 0 & 0\\ 0 & 5 & 4\\ 0 & 4 & 5\end{bmatrix}$

6406534366462. ✖ $\frac{1}{3}\begin{bmatrix}0 & 0 & 0\\ 0 & 2 & 1\\ 0 & 1 & 2\end{bmatrix}$

✔ **Official Answer Key: Option (B)**

MLT_132 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Label : Comprehension

Question Number : 7 Question Id : 6406531951440

Correct Marks : 3

Question Label : Short Answer Question

Find the covariance matrix C of the dataset. Enter the sum of all the entries of C as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_137 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 12 Question Id : 6406531951447

Correct Marks : 2

Question Label : Multiple Choice Question

If X is a data-matrix of shape $d \times n$, then X^TX and XX^T have the same eigenvectors.

Options :

6406536344688. ✖ True

6406536344689. ✔ False

Question Id : 6406531951448 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (13 to 14)

Question Label : Comprehension

Consider a dataset in $\mathbb{R}^2$ for a clustering problem with two clusters $(k = 2)$. The i^{th} data-point is represented as (x_i, y_i).

i	x_i	y_i
1	0	0
2	3	0
3	4	1
4	1	1

$z^{(t)} = (z_1^{(t)}, z_2^{(t)}, z_3^{(t)}, z_4^{(t)})$ indicates the cluster assignment vector at the end of time-step t. $z^{(0)}$ indicates the initial assignment. The cluster indices are represented by 1 and 2, that is, $z_i^{(t)} \in \{1, 2\}$.

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key: Option (B)

MLT_176 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 120 Question Id : 640653825113 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Kernel PCA with a polynomial kernel of degree 3 is run on a dataset of 100 data-points. Let ϕ be the transformation corresponding to this kernel. If the largest eigenvalue of the centered kernel matrix is 10, find the largest eigenvalue of the covariance matrix of the transformed dataset, where ϕ is used to effect the transformation.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.1

✔ **Official Answer Key (Numerical Value): 0.1**

MLT_216 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+20 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 154 Question Id : 6406531527890 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the covariance matrix of the dataset?

Options :

6406535115810. ✖ $\begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$

6406535115811. ✔ $\begin{pmatrix} 2.5 & 2 \\ 2 & 2.5 \end{pmatrix}$

6406535115812. ✖ $\begin{pmatrix} 3.5 & 2.5 \\ 2.5 & 3.5 \end{pmatrix}$

6406535115813. ✖ $\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$

✔ **Official Answer Key: Option (B)**

MLT_413 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 179 Question Id : 640653668622 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following matrices helps in transforming the eigenvectors of K to the eigenvectors of C?

Options :

6406532240077. ✔ X

6406532240078. ✖ X^T

6406532240079. ✖ C^{-1}

6406532240080. ✖ K^{-1}

6406532240081. ✖ X^{-1}

✔ Official Answer Key: Option (A)

MLT_414 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Covariance Matrix & Eigen-decomposition

Question Number : 180 Question Id : 640653668623 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose that the 95% of the variance of the data is captured using $d - 1$ eigenvectors. This implies that

Options :

6406532240082. ✖ The features are highly correlated (or dependent)

6406532240083. ✔ The features have low correlation (or independent)

✔ **Official Answer Key: Option (B)**

• **Reconstruction Error & Variance Captured** **(22 Questions)**

MLT_042 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 143 Question Id : 6406531298519 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Find the mean squared error for the training dataset.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

✔ **Official Answer Key (Numerical Value): 0**

MLT_210 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 148 Question Id : 6406531527883 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535115792. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535115793. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535115794. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535115795. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ **Official Answer Key: Option (B), Option (D)**

MLT_224 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 148 Question Id : 6406531528186 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535116635. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535116636. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535116637. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535116638. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_233 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 148 Question Id : 6406531528848 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535118566. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535118567. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535118568. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535118569. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_242 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 157 Question Id : 6406531525176 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535108206. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535108207. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535108208. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535108209. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_251 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 157 Question Id : 6406531525803 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535109983. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535109984. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535109985. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535109986. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_260 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 157 Question Id : 6406531526732 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535112628. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535112629. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535112630. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535112631. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_269 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 116 Question Id : 6406531535771 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535138021. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535138022. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535138023. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535138024. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_278 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 116 Question Id : 6406531536111 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535139121. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535139122. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535139123. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535139124. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_287 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 116 Question Id : 6406531536401 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535139952. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535139953. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535139954. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535139955. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_296 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 161 Question Id : 6406531533284 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535131074. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535131075. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535131076. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535131077. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_305 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 161 Question Id : 6406531533609 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535131970. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535131971. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535131972. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535131973. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_314 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 161 Question Id : 6406531533934 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535132866. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535132867. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535132868. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535132869. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_323 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 28 Question Id : 6406531534117 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535133291. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535133292. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535133293. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535133294. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_332 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 28 Question Id : 6406531534844 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535135319. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535135320. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535135321. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535135322. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_341 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 28 Question Id : 6406531534893 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535135407. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535135408. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535135409. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535135410. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_350 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 154 Question Id : 6406531527056 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535113505. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535113506. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535113507. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535113508. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_359 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 154 Question Id : 6406531527335 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535114263. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535114264. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535114265. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535114266. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_368 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 154 Question Id : 6406531527614 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535115021. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535115022. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535115023. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535115024. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_377 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 101 Question Id : 6406531531478 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535125923. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535125924. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535125925. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535125926. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_386 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 101 Question Id : 6406531532361 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535128415. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535128416. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535128417. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535128418. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ Official Answer Key: Option (B), Option (D)

MLT_395 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 1: Introduction; Unsupervised Learning - Representation learning - PCA › Reconstruction Error & Variance Captured

Question Number : 101 Question Id : 6406531532924 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of 5000 points in $\mathbb{R}^{50}$. If all these points lie in a 6-dimensional linear subspace, then which of the following will be true after applying the PCA?

Options :

6406535129985. ✖ There will be 50 non-zero eigenvalues of the covariance matrix.

6406535129986. ✔ There will be 6 non-zero eigenvalues of the covariance matrix.

6406535129987. ✖ Using the first 5 principal components, the dataset can be reconstructed with zero residue.

6406535129988. ✔ The variance captured by the top 6 principal components equals the total variance of the dataset.

✔ **Official Answer Key: Option (B), Option (D)**

Week 2: Unsupervised Learning - Representation learning - Kernel PCA (8 Questions)

- **Kernel Trick & Kernel Functions** (8 Questions)

MLT_055 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 181 Question Id : 640653470107 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

A dataset containing 200 examples in three-dimensional space has been transformed into a higher-dimensional space using a polynomial kernel of degree two. What will be the dimension of the transformed feature space?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Sub-Section Number :	5
Sub-Section Id :	64065367707
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653470108 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (182 to 183)

✔ **Official Answer Key (Numerical Value): 10**

MLT_071 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 183 Question Id : 640653737464 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a dataset consisting of 1000 samples in a 20-dimensional space. You apply Kernel PCA with a polynomial kernel of degree 3 to reduce the dimensionality of the data. Which of the following statements regarding Kernel PCA is true?

Options :

6406532468155. ✖ Kernel PCA always results in a lower-dimensional representation of the data compared to standard PCA.

6406532468156. ✖ Kernel PCA can only be applied to datasets that are linearly separable.

6406532468157. ✔ Choosing a higher degree polynomial kernel in Kernel PCA can lead to increased flexibility in capturing non-linear relationships.

6406532468158. ✖ None of these.

Sub-Section Number :	3
Sub-Section Id :	640653107659
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653737459 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (184 to 185)

Question Label : Comprehension

Standard PCA has been performed on a centered dataset in $\mathbb{R}^3$. The first two principal components are given below:

$$w_1 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad w_2 = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Consider the data point in the dataset: $[2 \quad 1 \quad -1]^T$. (a, b) is the representation of this point in the coordinate system formed by the two principal components given above. The first and second coordinates correspond to PC-1 and PC-2 respectively.

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key: Option (C)

MLT_129 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 4 Question Id : 6406531951436

Correct Marks : 3

Question Label : Short Answer Question

Consider a polynomial kernel of degree 3 on a feature space of dimension 4. How many features will the transformed space have$?$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

35

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_130 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 5 Question Id : 6406531951437

Correct Marks : 3

Question Label : Short Answer Question

Consider the Gaussian kernel, $k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ with $\sigma = 1$. Compute the value of $k((0,1),(1,0))$ and enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.34 to 0.40

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_136 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 11 Question Id : 6406531951446

Correct Marks : 2

Question Label : Multiple Choice Question

If $k_1 : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ and $k_2 : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ are valid kernels, then $2k_1 + 5k_2$ is also a valid kernel.

Options :

6406536344686. ✔ True

6406536344687. ✖ False

✔ Official Answer Key: Option (A)

MLT_174 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 118 Question Id : 640653825111 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$. Let k_1 and k_2 be two functions defined from $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$$
$$k_2(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2 + 1$$

Which of the following statements is true?

Options :

6406532775014. ✔ Both k_1 and k_2 are valid kernels.

6406532775015. ✖ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406532775016. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406532775017. ✖ Neither k_1 nor k_2 is a valid kernel.

Sub-Section Number :	4
Sub-Section Id :	640653122751
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_190 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 118 Question Id : 640653827580 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$. Let k_1 and k_2 be two functions defined from $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$$
$$k_2(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2 + 1$$

Which of the following statements is true?

Options :

6406532781538. ✔ Both k_1 and k_2 are valid kernels.

6406532781539. ✖ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406532781540. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406532781541. ✖ Neither k_1 nor k_2 is a valid kernel.

Sub-Section Number :	4
Sub-Section Id :	640653123502
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_200 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 2: Unsupervised Learning - Representation learning - Kernel PCA › Kernel Trick & Kernel Functions

Question Number : 118 Question Id : 640653827868 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$. Let k_1 and k_2 be two functions defined from $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$$
$$k_2(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2 + 1$$

Which of the following statements is true?

Options :

6406532782334. ✔ Both k_1 and k_2 are valid kernels.

6406532782335. ✖ k_1 is a valid kernel, but k_2 is not a valid kernel.

6406532782336. ✖ k_2 is a valid kernel, but k_1 is not a valid kernel.

6406532782337. ✖ Neither k_1 nor k_2 is a valid kernel.

Sub-Section Number :	4
Sub-Section Id :	640653123609
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means (72 Questions)

- **K-means Objective & Convergence** (24 Questions)

MLT_028 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 129 Question Id : 6406531298503 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

How does *K*-means++ algorithm enhance the initialization process as compared to the standard *K*-means algorithm?

Options :

6406534366427. ✖ It ensures the algorithm converges in a fixed number of iterations.

6406534366428. ✔ It improves centroid initialization by spreading them out, leading to better separated clusters.

6406534366429. ✖ It automatically calculates the best value of *K*.

6406534366430. ✖ It removes the need to run the algorithm multiple times.

✔ Official Answer Key: Option (B)

MLT_033 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 134 Question Id : 6406531298508 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are true?
Options :
6406534366447. ✖ EM algorithm converges to a global optimum.
6406534366448. ✔ Lloyd's algorithm converges to a local optimum.
6406534366449. ✖ Poor initialization does not affect the clusters produced by EM algorithm.
6406534366450. ✔ Poor initialization leads to a slower convergence of Lloyd's algorithm.
6406534366451. ✔ Lloyd's algorithm is sensitive to the number of clusters (k).

✔ Official Answer Key: Option (B), Option (D), Option (E)

MLT_034 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 135 Question Id : 6406531298509 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
For a linear regression problem, which among the following are true? Select all that apply.
Options :
6406534366452. ✔ Gradient descent will converge to a global minimum.
6406534366453. ✖ Gradient descent will converge to a local minimum (which is not a global minimum).
6406534366454. ✖ Stochastic gradient descent is guaranteed to converge to a global minimum.
6406534366455. ✔ Gradient descent will take lesser number of iterations to converge to a local minimum than stochastic gradient descent.

Sub-Section Number :	6
Sub-Section Id :	640653206767
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_052 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 178 Question Id : 640653470104 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Selectable Option : 0

Question Label : Multiple Select Question

With respect to the Lloyd's algorithm, choose the correct statements:

Options :

6406531562247. ✔ The partition configurations cannot repeat themselves.

6406531562248. ✖ After doing the reassignments (consider at least one point reassigned to the new cluster), we might get the same means for all clusters.

6406531562249. ✔ Objective function after making the re-assignments strictly reduces.

6406531562250. ✖ Objective function after making the re-assignments may increase.

6406531562251. ✔ A change in the objective function's value indicates that the partition configuration has changed.

6406531562252. ✖ For partitioning n data points across k partitions, Lloyd's algorithm takes k^n iterations to converge.

Sub-Section Number :	4
Sub-Section Id :	64065367706
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (C), Option (E)

MLT_074 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 186 Question Id : 640653737467 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Perform K-means clustering with $K = 2$ and initial cluster centers at (0, 0) and (6, 0). What are the final means of clusters after convergence?

Options :

6406532468163. ✖ (2, 0) and (8, 0)

6406532468164. ✖ (2, 0) and (8.5, 0)

6406532468165. ✖ (0, 0) and (8, 0)

6406532468166. ✔ (0, 0) and (8.5, 0)

6406532468167. ✖ None of these

✔ Official Answer Key: Option (D)

MLT_078 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 190 Question Id : 640653737465 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider Lloyd's algorithm used for k-means clustering and choose the correct statements:

Options :

6406532468159. ✔ K-means algorithm may get stuck at local minima.

6406532468160. ✖ It guarantees finding the optimal clustering (global minimum) in every run.

6406532468161. ✖ In practice, k should be as large as possible.

6406532468162. ✔ If the resources are limited and the data set is huge, it will be good to prefer K-means over K-means++.

✔ Official Answer Key: Option (A), Option (D)

MLT_085 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 190 Question Id : 640653737751 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider Lloyd's algorithm used for k-means clustering and choose the correct statements:

Options :

6406532468983. ✔ K-means algorithm may get stuck at local minima.

6406532468984. ✖ It guarantees finding the optimal clustering (global minimum) in every run.

6406532468985. ✖ In practice, k should be as large as possible.

6406532468986. ✔ If the resources are limited and the data set is huge, it will be good to prefer K-means over K-means++.

✔ Official Answer Key: Option (A), Option (D)

MLT_091 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 190 Question Id : 640653738037 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider Lloyd's algorithm used for k-means clustering and choose the correct statements:

Options :

6406532469807. ✔ K-means algorithm may get stuck at local minima.

6406532469808. ✖ It guarantees finding the optimal clustering (global minimum) in every run.

6406532469809. ✖ In practice, k should be as large as possible.

6406532469810. ✔ If the resources are limited and the data set is huge, it will be good to prefer K-means over K-means++.

✔ Official Answer Key: Option (A), Option (D)

MLT_127 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 2 Question Id : 6406531951434

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a clustering problem with 100 data-points and two clusters $(k = 2)$. As initialization, a subset of the dataset comprising the first fifty data-points, call this D_1, are assigned to cluster 1 and the remaining points, call this D_2, are assigned to cluster 2. K-means is run for this configuration and it converges after 10 iterations.
Consider the following scenarios:

Scenario-1: After convergence, it is observed that all points in D_1 are assigned to cluster 2 and all points in D_2 are assigned to cluster 1.

Scenario-2: After convergence, it is observed that all points in D_1 are assigned to cluster 1 and all points in D_2 are assigned to cluster 2.

Which of the following is true?

Options :

6406536344671. ✖ Scenario-1 is possible, but scenario-2 is impossible.

6406536344672. ✖ Scenario-1 is impossible, but scenario-2 is possible.

6406536344673. ✖ Both scenarios are possible.

6406536344674. ✔ Both scenarios are impossible.

Sub-Section Number :	3
Sub-Section Id :	640653360303
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (D)

MLT_131 | [MCQ] | Marks: +3.0 | *MLT (1)*

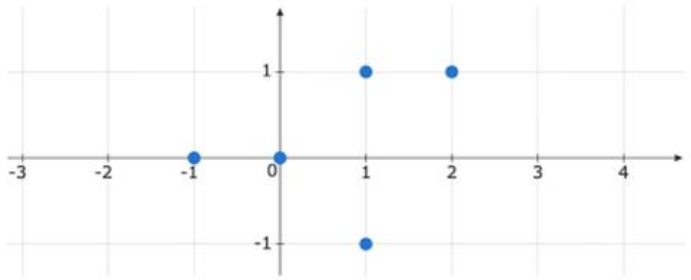
Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 6 Question Id : 6406531951438

Correct Marks : 3

Question Label : Short Answer Question

Consider the following data-points in a clustering problem with five data-points and three clusters (k = 3):



K-means^{++} is used to initialize the three means. The first mean is chosen to be $\mu_1 = (0,0)$ and the second mean to be $\mu_2 = (1,1)$. Find the probability of choosing $(1,-1)$ as the third mean. That is, compute $P(\mu_3 = (1,-1) \mid \mu_1 = (0,0), \mu_2 = (1,1))$. Enter your answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Sub-Section Number :	5
Sub-Section Id :	640653360305
Question Shuffling Allowed :	No

Question Id : 6406531951439 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (7 to 8)

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

MLT_138 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 13 Question Id : 6406531951449

Correct Marks : 2

Question Label : Multiple Choice Question

If $z^{(0)} = (1,1,2,2)$, which of the following is the final cluster assignment after running K-means on this dataset with this initial configuration?

Options :

6406536344690. ✖ $(1,1,2,2)$

6406536344691. ✔ $(1,2,2,1)$

6406536344692. ✖ $(1,2,1,2)$

6406536344693. ✖ $(1,1,1,2)$

6406536344694. ✖ $(1,1,2,1)$

✔ Official Answer Key: Option (B)

MLT_158 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 134 Question Id : 6406531297968 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is/are true?

Options :

6406534364905. ✖ EM algorithm converges to a global optimum.

6406534364906. ✔ Lloyd's algorithm converges to a local optimum.

6406534364907. ✖ Poor initialization does not affect the clusters produced by EM algorithm.

6406534364908. ✔ Poor initialization leads to a slower convergence of Lloyd's algorithm.

6406534364909. ✔ Lloyd's algorithm is sensitive to the number of clusters (k).

✔ Official Answer Key: Option (B), Option (D), Option (E)

MLT_159 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 135 Question Id : 6406531297969 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
For a linear regression problem, which among the following are true? Select all that apply.
Options :
6406534364910. ✔ Gradient descent will converge to a global minimum.
6406534364911. ✖ Gradient descent will converge to a local minimum (which is not a global minimum).
6406534364912. ✖ Stochastic gradient descent is guaranteed to converge to a global minimum.
6406534364913. ✔ Gradient descent will take lesser number of iterations to converge to a local minimum than stochastic gradient descent.

Sub-Section Number :	6
Sub-Section Id :	640653206565
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_167 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 134 Question Id : 6406531298238 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are true?
Options :
6406534365676. ✖ EM algorithm converges to a global optimum.
6406534365677. ✔ Lloyd's algorithm converges to a local optimum.
6406534365678. ✖ Poor initialization does not affect the clusters produced by EM algorithm.
6406534365679. ✔ Poor initialization leads to a slower convergence of Lloyd's algorithm.
6406534365680. ✔ Lloyd's algorithm is sensitive to the number of clusters (k).

✔ Official Answer Key: Option (B), Option (D), Option (E)

MLT_168 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 135 Question Id : 6406531298239 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

For a linear regression problem, which among the following are true? Select all that apply.

Options :

6406534365681. ✔ Gradient descent will converge to a global minimum.

6406534365682. ✖ Gradient descent will converge to a local minimum (which is not a global minimum).

6406534365683. ✖ Stochastic gradient descent is guaranteed to converge to a global minimum.

6406534365684. ✔ Gradient descent will take lesser number of iterations to converge to a local minimum than stochastic gradient descent.

Sub-Section Number :	6
Sub-Section Id :	640653206666
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_183 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 127 Question Id : 640653825122 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If (x_1, y_1) is the mean of the first cluster after convergence, find the value of $x_1 + y_1$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-5

✔ Official Answer Key (Numerical Value): -5

MLT_184 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 128 Question Id : 640653825123 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If (x_2, y_2) is the mean of the second cluster after convergence, find the value of $x_2 + y_2$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Sub-Section Number :	9
Sub-Section Id :	640653122756
Question Shuffling Allowed :	No

Question Id : 640653825127 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (129 to 130)

Question Label : Comprehension

In the context of Bayesian estimation, consider a Beta prior for the parameter p of a Bernoulli distribution:

$$p \sim \text{Beta}(4, 3)$$

The dataset has 8 ones and 5 zeros.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 9**

MLT_196 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 128 Question Id : 640653827592 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
If (x_2, y_2) is the mean of the second cluster after convergence, find the value of $x_2 + y_2$.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
9

Sub-Section Number :	9
Sub-Section Id :	640653123507
Question Shuffling Allowed :	No

Question Id : 640653827596 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (129 to 130)
Question Label : Comprehension
In the context of Bayesian estimation, consider a Beta prior for the parameter p of a Bernoulli distribution:

$$p \sim \text{Beta}(4, 3)$$

The dataset has 8 ones and 5 zeros.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key (Numerical Value): 9**

MLT_206 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 128 Question Id : 640653827880 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If (x_2, y_2) is the mean of the second cluster after convergence, find the value of $x_2 + y_2$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Sub-Section Number :	9
Sub-Section Id :	640653123614
Question Shuffling Allowed :	No

Question Id : 640653827884 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (129 to 130)

Question Label : Comprehension

In the context of Bayesian estimation, consider a Beta prior for the parameter p of a Bernoulli distribution:

$$p \sim \text{Beta}(4, 3)$$

The dataset has 8 ones and 5 zeros.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 9**

MLT_221 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+20 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 159 Question Id : 6406531527894 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Perform *K*-means clustering with $K = 2$ and initial cluster centers at (−2, 3) and (2,−1). What are the final means of clusters after convergence?

Options :

6406535115816. ✔ $(-1.75, 2.5)$ and $(1.5, -1.5)$

6406535115817. ✖ $(-2, 3)$ and $(2, -1)$

6406535115818. ✖ $(-1.33, 3.66)$ and $(0, -1.33)$

6406535115819. ✖ $(1.33, 1)$ and $(2.66, 2)$

✔ **Official Answer Key: Option (A)**

MLT_222 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+20 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 160 Question Id : 6406531527895 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Suppose we introduce a new data point (0, 0) to the existing dataset and perform *K* means with the same initial cluster centers. How many iterations would it require now for convergence?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

✔ **Official Answer Key (Numerical Value): 1**

MLT_404 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 169 Question Id : 640653668610 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider 9 data points $x_1, x_2, \ldots, x_9$: (2,3),(1,2),(3,1),(5,6),(7,8),(9,9),(6,7),(10,10),(11,11). Suppose we aim to form 3 clusters from these points using the K-Means algorithm. After the first iteration, clusters C_1, C_2, C_3 have the following data points:

C_1: (1, 2), (3, 1)

C_2: (2, 3), (5, 6), (6,7)

C_3: (9, 9), (7, 8), (10, 10), (11, 11)

After the second iteration, to which cluster will the data point (2, 3) be reassigned?

Options :

6406532240051. ✔ C_1

6406532240052. ✖ C_2

6406532240053. ✖ C_3

6406532240054. ✖ It remains in the same cluster as in the first iteration

Sub-Section Number :	3
Sub-Section Id :	64065395191
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLT_416 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 169 Question Id : 640653668971 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider 9 data points $x_1, x_2, \ldots, x_9$: (2,3),(1,2),(3,1),(5,6),(7,8),(9,9),(6,7),(10,10),(11,11). Suppose we aim to form 3 clusters from these points using the K-Means algorithm. After the first iteration, clusters C_1, C_2, C_3 have the following data points:

C_1: (1, 2), (3, 1)

C_2: (2, 3), (5, 6), (6,7)

C_3: (9, 9), (7, 8), (10, 10), (11, 11)

After the second iteration, to which cluster will the data point (2, 3) be reassigned?

Options :

6406532241147. ✔ C_1

6406532241148. ✖ C_2

6406532241149. ✖ C_3

6406532241150. ✖ It remains in the same cluster as in the first iteration

Sub-Section Number :	3
Sub-Section Id :	64065395296
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLT_424 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › K-means Objective & Convergence

Question Number : 169 Question Id : 640653669232 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider 9 data points $x_1, x_2, \ldots, x_9$: (2,3),(1,2),(3,1),(5,6),(7,8),(9,9),(6,7),(10,10),(11,11). Suppose we aim to form 3 clusters from these points using the K-Means algorithm. After the first iteration, clusters C_1, C_2, C_3 have the following data points:

C_1: (1, 2), (3, 1)

C_2: (2, 3), (5, 6), (6,7)

C_3: (9, 9), (7, 8), (10, 10), (11, 11)

After the second iteration, to which cluster will the data point (2, 3) be reassigned?

Options :

6406532241899. ✔ C_1

6406532241900. ✖ C_2

6406532241901. ✖ C_3

6406532241902. ✖ It remains in the same cluster as in the first iteration

Sub-Section Number :	3
Sub-Section Id :	64065395395
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

• Cluster Centers & Distance Metrics **(27 Questions)**

MLT_064 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 190 Question Id : 640653470119 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

What will be the final cluster means?

Options :

6406531562268. ✖ $\mu_1 = -15,\ \mu_2 = 0,\ \mu_3 = 5$

6406531562269. ✖ $\mu_1 = -10,\ \mu_2 = 0,\ \mu_3 = 15$

6406531562270. ✖ $\mu_1 = -12.5,\ \mu_2 = 0,\ \mu_3 = 16.25$

6406531562271. ✖ $\mu_1 = -12.5,\ \mu_2 = 2.5,\ \mu_3 = 16.25$

6406531562272. ✔ $\mu_1 = -12.5,\ \mu_2 = 2.5,\ \mu_3 = 20$

Sub-Section Number :	8
Sub-Section Id :	64065367710
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653470120 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (191 to 192)

✔ Official Answer Key: Option (E)

MLT_065 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23 (+3 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 191 Question Id : 640653470121 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Which point has the highest probability of being chosen as the 2^{nd} cluster mean? Use the manhattan distance to compute the distances.

Options :

6406531562273. ✖ (3,1)

6406531562274. ✔ (4,7)

6406531562275. ✖ (-1,9)

6406531562276. ✖ (4,-2)

✔ Official Answer Key: Option (B)

MLT_066 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 192 Question Id : 640653470122 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

If the point with the highest score is chosen as the 2^{nd} cluster mean(answer from previous question), Which point has the highest probability of being chosen as 3^{rd} the cluster mean? Use the manhattan distance to compute the distances.

Options :

6406531562277. ✖ (3,1)

6406531562278. ✖ (4,7)

6406531562279. ✔ (-1,9)

6406531562280. ✖ (4,-2)

✔ Official Answer Key: Option (C)

MLT_075 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 187 Question Id : 640653737468 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

After introducing a new data point (7, 0) , the cluster centers were updated. Enter the sum of the updated x-coordinates for both cluster centers accurately, rounding your answer to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

8.18 to 8.22

Sub-Section Number :	4
Sub-Section Id :	640653107660
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653737470 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (188 to 189)

✔ **Official Answer Key (Numerical Value): 8.18 to 8.22**

MLT_211 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 149 Question Id : 6406531527885 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535115800. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535115801. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535115802. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535115803. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_225 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 149 Question Id : 6406531528188 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535116643. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535116644. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535116645. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535116646. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_234 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 149 Question Id : 6406531528850 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535118574. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535118575. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535118576. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535118577. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_243 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 158 Question Id : 6406531525178 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535108214. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535108215. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535108216. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535108217. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_252 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 158 Question Id : 6406531525805 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535109991. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535109992. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535109993. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535109994. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_261 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 158 Question Id : 6406531526734 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535112636. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535112637. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535112638. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535112639. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_270 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 117 Question Id : 6406531535773 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535138029. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535138030. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535138031. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535138032. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_279 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 117 Question Id : 6406531536113 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535139129. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535139130. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535139131. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535139132. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_288 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 117 Question Id : 6406531536403 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535139960. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535139961. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535139962. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535139963. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_297 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 162 Question Id : 6406531533286 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535131082. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535131083. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535131084. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535131085. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_306 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 162 Question Id : 6406531533611 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535131978. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535131979. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535131980. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535131981. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_315 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 162 Question Id : 6406531533936 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535132874. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535132875. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535132876. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535132877. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_324 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 29 Question Id : 6406531534119 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535133299. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535133300. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535133301. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535133302. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_333 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 29 Question Id : 6406531534846 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535135327. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535135328. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535135329. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535135330. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_342 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 29 Question Id : 6406531534895 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535135415. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535135416. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535135417. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535135418. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_351 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 155 Question Id : 6406531527058 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535113513. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535113514. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535113515. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535113516. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_360 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 155 Question Id : 6406531527337 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535114271. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535114272. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535114273. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535114274. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_369 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 155 Question Id : 6406531527616 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535115029. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535115030. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535115031. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535115032. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_378 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 102 Question Id : 6406531531480 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535125931. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535125932. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535125933. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535125934. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_387 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 102 Question Id : 6406531532363 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535128423. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535128424. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535128425. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535128426. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_396 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 102 Question Id : 6406531532926 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the dataset consisting of the following points:

$$(0,0),\ (0,2),\ (2,0),\ (2,2).$$

We want to run the K-means++ algorithm with $K = 2$. Suppose the first cluster center chosen is $(0,0)$. Which of the following statements are correct?

Hint: The score of a datapoint x is given by $S(x) = \min_{j=1,2,\ldots,l-1}(||x - \mu_j^0||^2)$, where μ_j^0 is the initial cluster center.

Options :

6406535129993. ✖ The points (0, 2), (2, 0), and (2, 2) have the equal probability of being chosen as the second cluster center.

6406535129994. ✔ The probability of selecting (2, 2) as the second cluster center is twice that of selecting (0, 2).

6406535129995. ✔ If (2, 2) is chosen as the second cluster center, the algorithm will always converge in one iteration.

6406535129996. ✔ The tie-breaking rule will affect the final cluster centers.

✔ Official Answer Key: Option (B), Option (C), Option (D)

MLT_411 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 177 Question Id : 640653668613 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let's assume that (3, 3), (10, 10), and (12, 12) are the cluster means. You introduce a new data point, (−5, 10). Which cluster mean is the data point (−5, 10) closest to when using the Euclidean distance method?

Options :

6406532240059. ✖ (3, 3)

6406532240060. ✔ (10, 10)

6406532240061. ✖ (12, 12)

6406532240062. ✖ None of these

Sub-Section Number :	7
Sub-Section Id :	64065395195
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653668620 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (178 to 180)

✔ Official Answer Key: Option (B)

MLT_442 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*
Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Cluster Centers & Distance Metrics

Question Number : 27 Question Id : 640653995447 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question
Which point has the lowest probability of being chosen as the 3rd cluster mean? Use squared distance to calculate the scores.
Options :
6406533360638. ✖ (1,0)
6406533360639. ✔ (2,3)
6406533360640. ✖ (3,2)
6406533360641. ✖ (1,2)

✔ **Official Answer Key: Option (B)**

• **Initialization & K-means++** **(21 Questions)**

MLT_209 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 147 Question Id : 6406531527882 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535115788. ✔ It always converges.
6406535115789. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535115790. ✔ It is sensitive to outliers.
6406535115791. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244938
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A), Option (C)**

MLT_223 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 147 Question Id : 6406531528185 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535116631. ✔ It always converges.

6406535116632. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535116633. ✔ It is sensitive to outliers.

6406535116634. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653245034
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_232 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 147 Question Id : 6406531528847 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535118562. ✔ It always converges.

6406535118563. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535118564. ✔ It is sensitive to outliers.

6406535118565. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653245219
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_241 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 156 Question Id : 6406531525175 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535108202. ✔ It always converges.
6406535108203. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535108204. ✔ It is sensitive to outliers.
6406535108205. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244051
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_250 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 156 Question Id : 6406531525802 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535109979. ✔ It always converges.
6406535109980. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535109981. ✔ It is sensitive to outliers.
6406535109982. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244244
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_259 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 156 Question Id : 6406531526731 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535112624. ✔ It always converges.
6406535112625. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535112626. ✔ It is sensitive to outliers.
6406535112627. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244535
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_268 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 115 Question Id : 6406531535770 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535138017. ✔ It always converges.
6406535138018. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535138019. ✔ It is sensitive to outliers.
6406535138020. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653247423
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_277 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 115 Question Id : 6406531536110 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535139117. ✔ It always converges.
6406535139118. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535139119. ✔ It is sensitive to outliers.
6406535139120. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653247525
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_286 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 115 Question Id : 6406531536400 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535139948. ✔ It always converges.
6406535139949. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535139950. ✔ It is sensitive to outliers.
6406535139951. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653247627
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_295 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 160 Question Id : 6406531533283 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535131070. ✔ It always converges.
6406535131071. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535131072. ✔ It is sensitive to outliers.
6406535131073. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246686
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_304 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 160 Question Id : 6406531533608 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535131966. ✔ It always converges.
6406535131967. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535131968. ✔ It is sensitive to outliers.
6406535131969. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246782
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_313 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 160 Question Id : 6406531533933 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535132862. ✔ It always converges.
6406535132863. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535132864. ✔ It is sensitive to outliers.
6406535132865. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246878
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_322 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 27 Question Id : 6406531534116 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535133287. ✔ It always converges.
6406535133288. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535133289. ✔ It is sensitive to outliers.
6406535133290. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246924
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_331 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 27 Question Id : 6406531534843 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535135315. ✔ It always converges.
6406535135316. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535135317. ✔ It is sensitive to outliers.
6406535135318. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653247142
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_340 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 27 Question Id : 6406531534892 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535135403. ✔ It always converges.
6406535135404. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535135405. ✔ It is sensitive to outliers.
6406535135406. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653247156
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_349 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 153 Question Id : 6406531527055 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535113501. ✔ It always converges.

6406535113502. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535113503. ✔ It is sensitive to outliers.

6406535113504. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244639
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_358 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 153 Question Id : 6406531527334 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535114259. ✔ It always converges.

6406535114260. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535114261. ✔ It is sensitive to outliers.

6406535114262. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244743
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_367 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 153 Question Id : 6406531527613 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535115017. ✔ It always converges.

6406535115018. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535115019. ✔ It is sensitive to outliers.

6406535115020. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653244847
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_376 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 100 Question Id : 6406531531477 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which among the following is true with respect to the Lloyds' algorithm?

Options :

6406535125919. ✔ It always converges.

6406535125920. ✖ It may fail to converge for an initialization while it may converge for other initialization.

6406535125921. ✔ It is sensitive to outliers.

6406535125922. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246078
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_385 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 100 Question Id : 6406531532360 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535128411. ✔ It always converges.
6406535128412. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535128413. ✔ It is sensitive to outliers.
6406535128414. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246371
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

MLT_394 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 3: Unsupervised Learning - Clustering - K-means/Kernel K-means › Initialization & K-means++

Question Number : 100 Question Id : 6406531532923 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which among the following is true with respect to the Lloyds' algorithm?
Options :
6406535129981. ✔ It always converges.
6406535129982. ✖ It may fail to converge for an initialization while it may converge for other initialization.
6406535129983. ✔ It is sensitive to outliers.
6406535129984. ✖ The solution produced is always the optimal.

Sub-Section Number :	3
Sub-Section Id :	640653246566
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm (39 Questions)

• Maximum Likelihood Estimation (MLE) (10 Questions)

MLT_016 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 185 Question Id : 640653345316 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider following label vector for a movie data $\mathrm{D} = (\mathrm{X}, \mathrm{y})$

$$\mathrm{X} = \begin{bmatrix} 2019 & \text{'Uri: The Surgical Strike'} & 8.3 \\ 2009 & \text{'3 Idiots'} & 8.4 \end{bmatrix}$$

$$\mathrm{y} = \begin{bmatrix} \text{'Drama', 'Action', 'War'} \\ \text{'Comedy', 'Drama'} \end{bmatrix}$$

X has year of release, name of the movie and IMDb rating as features. And y has corresponding genres. Predicting genres of a movie is following type of problem:

Options :

6406531148563. ✔ Multi-label, multi-class classification problem

6406531148564. ✖ Single-label, multi-class classification problem

6406531148565. ✖ Single-label, binary class classification problem

6406531148566. ✖ Multi-label, binary class classification problem

✔ Official Answer Key: Option (A)

MLT_020 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 189 Question Id : 640653345319 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

Which of the following correctly computes the solution of linear regression problem via normal equation method? Assume necessary imports.

Options :

6406531148575. ✔
```
w = np.linalg.pinv(X)@y
```

6406531148576. ✖
```
w = X@X.T@y
```

6406531148577. ✖
```
w = X.T@np.linalg.inv(X)@y
```

6406531148578. ✖
```
w = X.T@np.linalg.pinv(X@y)@y
```

6406531148579. ✖
```
w = np.linalg.inv(X.T@X)(X.T@y)
```

✔ **Official Answer Key: Option (A)**

MLT_024 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 193 Question Id : 640653345335 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

A classification model was trained on some training examples with two features and three classes (0, 1 and 2) using least square algorithm. The learnt weight matrix is

$$\mathbf{W} = \begin{bmatrix} -1.2 & 0.6 & 0.5 \\ 0.3 & 0.8 & -1.5 \\ 0.4 & 1.3 & 1 \end{bmatrix}$$

(Remember for k class classification problem with m features, shape of weight matrix is $(m + 1) \times k$)

Which class will the sample (-1, 1) be classified to?

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Sub-Section Number :	8
Sub-Section Id :	64065348801
Question Shuffling Allowed :	Yes

✔ Official Answer Key (Numerical Value): 2

MLT_076 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 188 Question Id : 640653737471 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What is the log-likelihood of this dataset under this distribution? ln represents the natural logarithm or $\log_e$.

Options :

6406532468170. ✖ $\prod_{i=1}^{6} \lambda e^{-\lambda x_i}$

6406532468171. ✖ $\sum_{i=1}^{6} \lambda e^{-\lambda x_i}$

6406532468172. ✖ $\prod_{i=1}^{6} [\ln(\lambda) - \lambda x_i]$

6406532468173. ✔

$\sum_{i=1}^{6} [\ln(\lambda) - \lambda x_i]$

✔ **Official Answer Key: Option (D)**

MLT_080 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+5 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 192 Question Id : 640653737469 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset with 100 total data points. Each data point is classified as either type A or type B. We model this using a Bernoulli distribution, where p is the probability of a data point being type A. If the maximum likelihood estimate (MLE) of p based on the dataset is 0.4, how many data points of type B are there in this dataset?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

60

✔ Official Answer Key (Numerical Value): 60

MLT_101 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 140 Question Id : 6406531113835 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose you toss a coin four times and observe no heads. You then give the coin to your friend, who tosses it until the first head occurs. Your friend tosses the coin a total of five times. Let p denote the probability that the coin comes up heads. Find the maximum likelihood estimate of p.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.10 to 0.12

Sub-Section Number :	7
Sub-Section Id :	640653168536
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 0.10 to 0.12**

MLT_139 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 15 Question Id : 6406531951452

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following expressions correctly represents the log-likelihood function $\ell(\theta)$ for the given data?

Options :

6406536344700. ✖ $\sum_{i=1}^{n} [\theta \log f_1(x_i) + (1-\theta) \log f_2(x_i)]$

6406536344701. ✖ $\log \left(\sum_{i=1}^{n} [\theta f_1(x_i) + (1-\theta) f_2(x_i)] \right)$

6406536344702. ✔ $\sum_{i=1}^{n} \log \left(\theta e^{-x_i} + (1-\theta) 3e^{-3x_i} \right)$

6406536344703. ✖ $\sum_{i=1}^{n} [\log \theta + \log f_1(x_i)]$

✔ **Official Answer Key: Option (C)**

MLT_187 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 131 Question Id : 640653825125 Question Type : MCQ

Correct Marks : 2.5

Question Label : Multiple Choice Question

Find the log-likelihood of the dataset under this distribution.

Options :

6406532775035. ✔ $\sum_{i=1}^{n} [-\lambda + x_i \log \lambda - \log (x_i!)]$

6406532775036. ✖ $\prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$

6406532775037. ✖ $\sum_{i=1}^{n} [-\lambda + \lambda \log x_i - \log (x_i!)]$

6406532775038. ✖ $\prod_{i=1}^{n} [-\lambda + x_i \log \lambda - \log (x_i!)]$

✔ Official Answer Key: Option (A)

MLT_188 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 132 Question Id : 640653825126 Question Type : SA

Correct Marks : 2.5

Question Label : Short Answer Question

Consider a dataset that has 25 data-points. The data-point x_i and its frequency is given in the following table:

x_i	Frequency
0	1
1	4
2	6
3	9
4	5

In case the table is not clear: the value 0 appears once in the dataset, the value 1 appears four times in the dataset, and so on. Find the maximum likelihood estimate for the parameter λ of the Poisson distribution given this dataset.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2.52

✔ **Official Answer Key (Numerical Value): 2.52**

MLT_406 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Maximum Likelihood Estimation (MLE)

Question Number : 171 Question Id : 640653668614 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider a dataset that has 200 data-points, each of which is either a zero or one. The Bernoulli distribution is used to model this data. If the MLE estimate for the parameter p of the Bernoulli distribution is 0.3, how many zeros does the dataset have?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

140

✔ **Official Answer Key (Numerical Value): 140**

- **Expectation-Maximization (EM) Algorithm** **(2 Questions)**

MLT_032 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Expectation-Maximization (EM) Algorithm

Question Number : 133 Question Id : 6406531298507 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You are presented with a dataset that has latent variables that influences your data. You are asked to use Expectation-Maximization algorithm to best capture the data. How would you define the steps in the EM (Expectation and Maximization) algorithm?

Options :

6406534366443. ✔ E-step: Estimate the latent variables in the dataset, M-step: Maximize the likelihood over the model parameters.

6406534366444. ✖ E-step: Estimate the number of latent variables in the dataset, M-step: Maximize the likelihood over the model parameters.

6406534366445. ✖ E-step: Estimate the likelihood over the model parameters, M-step: Maximize the number of latent variables in the dataset.

6406534366446. ✖ E-step: Estimate the likelihood over the model parameters, M-step: Maximize the number of parameters in the model.

Sub-Section Number :	5
Sub-Section Id :	640653206766
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLT_140 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Expectation-Maximization (EM) Algorithm

Question Number : 16 Question Id : 6406531951453

Correct Marks : 3

Question Label : Short Answer Question

Consider the E-step of the EM algorithm. Introduce latent variables $Z_i \in \{0,1\}$ such that $Z_i = 1$ if X_i is generated from f_1. Compute the posterior responsibility

$$\gamma_i^{(0)} = P(Z_i = 1 \mid X_i = x_i, \theta^{(0)}) = \frac{\theta^{(0)} f_1(x_i)}{\theta^{(0)} f_1(x_i) + (1 - \theta^{(0)}) f_2(x_i)}$$

for each observation x_i. Write down $\sum_{i=1}^{n} \gamma_i^{(0)}$ as your answer. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.74 to 2.80

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

- **Bayesian Estimation & Priors** **(27 Questions)**

MLT_105 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 144 Question Id : 6406531113841 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct options from the following:

Options :

6406533774423. ✖ Posterior distribution for p is Beta(91, 11).

6406533774424. ✔ Posterior distribution for p is Beta(11, 91).

6406533774425. ✖ Posterior distribution for p is Bernoulli(0.1).

6406533774426. ✖ Posterior distribution for p is Bernoulli(0.9).

✔ Official Answer Key: Option (B)

MLT_106 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 145 Question Id : 6406531113842 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Find the posterior mean. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.10 to 0.11

✔ Official Answer Key (Numerical Value): 0.10 to 0.11

MLT_141 | [MCQ] | Marks: +3.0 | *MLT (1)*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 17 Question Id : 6406531951454

Correct Marks : 3

Question Label : Multiple Choice Question

Using a Maximum A Posteriori (MAP) decision rule, a data point x_i is assigned to mixture 2 if the posterior responsibility $\gamma_i^{(0)} \leq 0.5$. Based on the values calculated in the E-step, which data point(s) are classified as belonging to Component 2?

Options :

6406536344705. ✔ $\{0.2\}$ only

6406536344706. ✖ $\{0.2, 0.5\}$ only

6406536344707. ✖ $\{1.1, 1.8\}$ only

6406536344708. ✖ All four points

✔ Official Answer Key: Option (A)

MLT_185 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 129 Question Id : 640653825128 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the posterior?

Options :

6406532775040. ✔ Beta(12, 8)

6406532775041. ✖ Beta(11, 7)

6406532775042. ✖ *Br*(0.5)

6406532775043. ✖ Beta(8, 5)

✔ Official Answer Key: Option (A)

MLT_197 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 129 Question Id : 640653827597 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the posterior?

Options :

6406532781564. ✔ Beta(12, 8)

6406532781565. ✖ Beta(11, 7)

6406532781566. ✖ *Br*(0.5)

6406532781567. ✖ Beta(8, 5)

✔ Official Answer Key: Option (A)

MLT_207 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 129 Question Id : 640653827885 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the posterior?

Options :

6406532782360. ✔ Beta(12, 8)

6406532782361. ✖ Beta(11, 7)

6406532782362. ✖ *Br*(0.5)

6406532782363. ✖ Beta(8, 5)

✔ Official Answer Key: Option (A)

MLT_212 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 150 Question Id : 6406531527886 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535115804. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535115805. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535115806. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535115807. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244939
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_226 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 150 Question Id : 6406531528189 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535116647. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535116648. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535116649. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535116650. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653245035
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_235 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 150 Question Id : 6406531528851 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535118578. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535118579. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535118580. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535118581. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653245220
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_244 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 159 Question Id : 6406531525179 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535108218. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535108219. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535108220. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535108221. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244052
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_253 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 159 Question Id : 6406531525806 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535109995. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535109996. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535109997. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535109998. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244245
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_262 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 159 Question Id : 6406531526735 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535112640. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535112641. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535112642. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535112643. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244536
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_271 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 118 Question Id : 6406531535774 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535138033. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535138034. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535138035. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535138036. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653247424
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_280 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 118 Question Id : 6406531536114 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535139133. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535139134. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535139135. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535139136. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653247526
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_289 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 118 Question Id : 6406531536404 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535139964. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535139965. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535139966. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535139967. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653247628
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_298 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 163 Question Id : 6406531533287 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535131086. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535131087. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535131088. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535131089. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246687
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_307 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 163 Question Id : 6406531533612 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535131982. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535131983. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535131984. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535131985. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246783
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_316 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 163 Question Id : 6406531533937 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535132878. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535132879. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535132880. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535132881. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246879
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_325 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 30 Question Id : 6406531534120 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535133303. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535133304. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535133305. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535133306. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246925
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_334 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 30 Question Id : 6406531534847 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535135331. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535135332. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535135333. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535135334. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653247143
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_343 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIE4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 30 Question Id : 6406531534896 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535135419. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535135420. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535135421. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535135422. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653247157
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_352 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 156 Question Id : 6406531527059 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535113517. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535113518. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535113519. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535113520. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244640
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_361 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 156 Question Id : 6406531527338 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535114275. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535114276. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535114277. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535114278. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244744
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_370 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 156 Question Id : 6406531527617 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535115033. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535115034. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535115035. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535115036. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653244848
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_379 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 103 Question Id : 6406531531481 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535125935. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535125936. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535125937. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535125938. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246079
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_388 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 103 Question Id : 6406531532364 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535128427. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535128428. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535128429. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535128430. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246372
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)

MLT_397 | [MSQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025*

Category: Week 4: Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm › Bayesian Estimation & Priors

Question Number : 103 Question Id : 6406531532927 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose that 100 items are sampled from a manufacturing process, where three items are found to be defective. Consider Beta(a, b) as a prior distribution for the proportion θ of defective items with the following parameters:

- $a = 1, b = 1$
- $a = 0.5, b = 0.5$

If posterior mean is used as the estimate for θ, then which of the following is correct?

Hint: Expectation of a Beta(a, b) is given by $\frac{a}{(a+b)}$.

Options :

6406535129997. ✔ Posterior mean for Beta(1, 1) is 0.039.

6406535129998. ✖ Posterior mean for Beta(1, 1) is 0.03.

6406535129999. ✖ Posterior mean for Beta(0.5, 0.5) is 0.025.

6406535130000. ✔ Posterior mean for Beta(0.5, 0.5) is 0.034.

Sub-Section Number :	4
Sub-Section Id :	640653246567
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D)